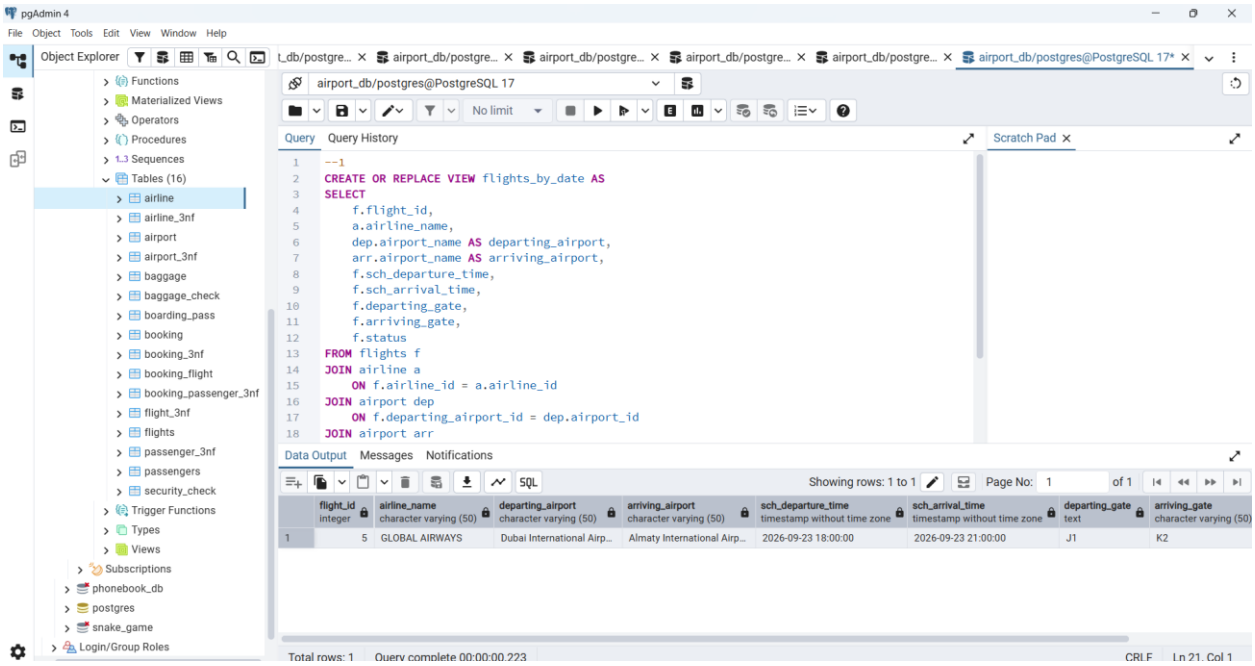
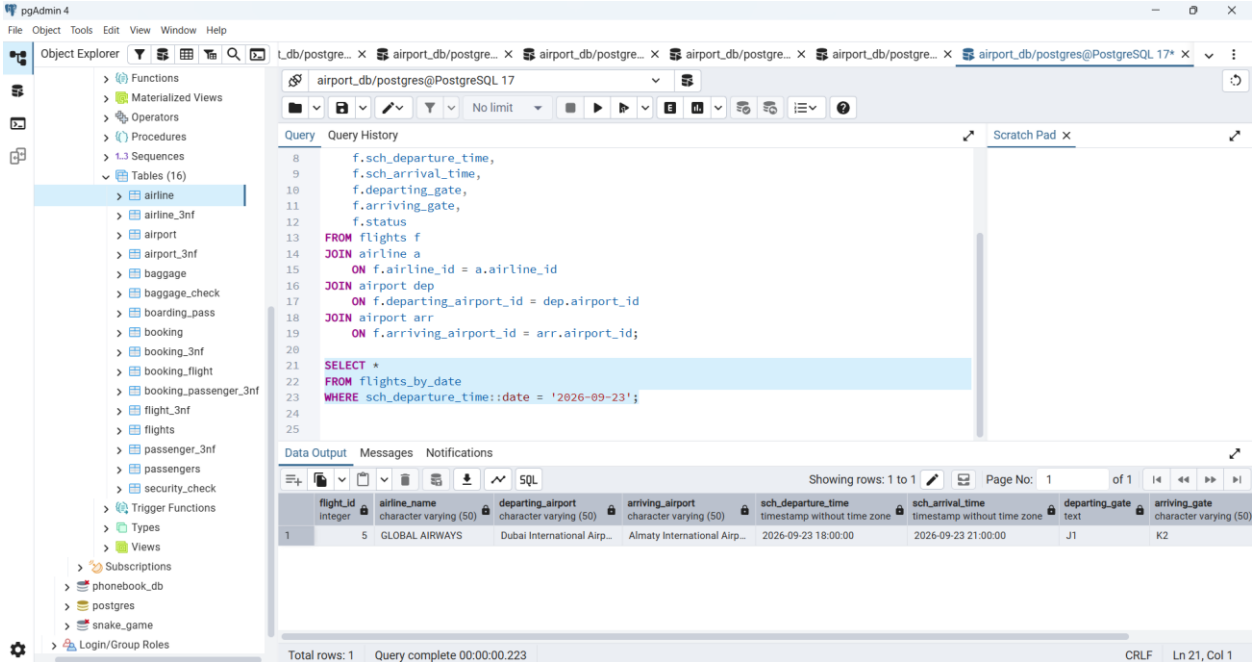
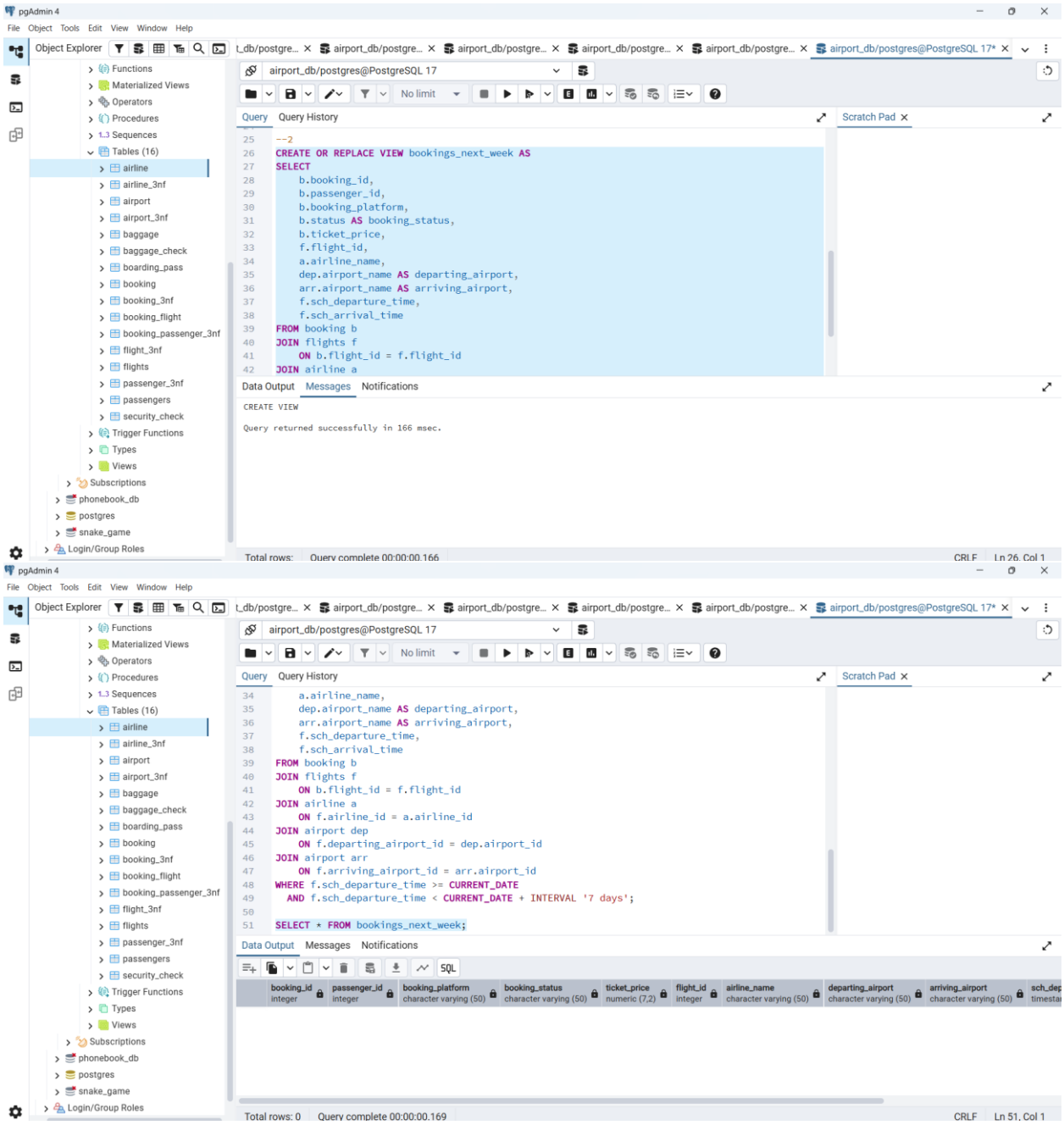
Laboratory work 8

1



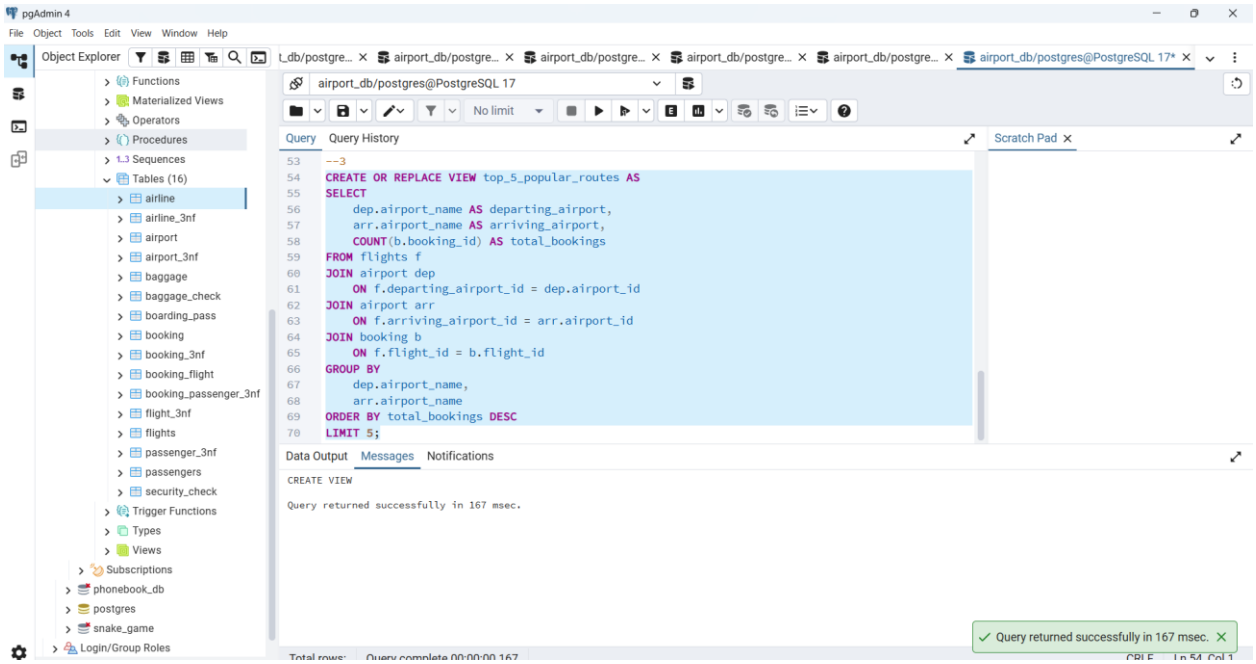


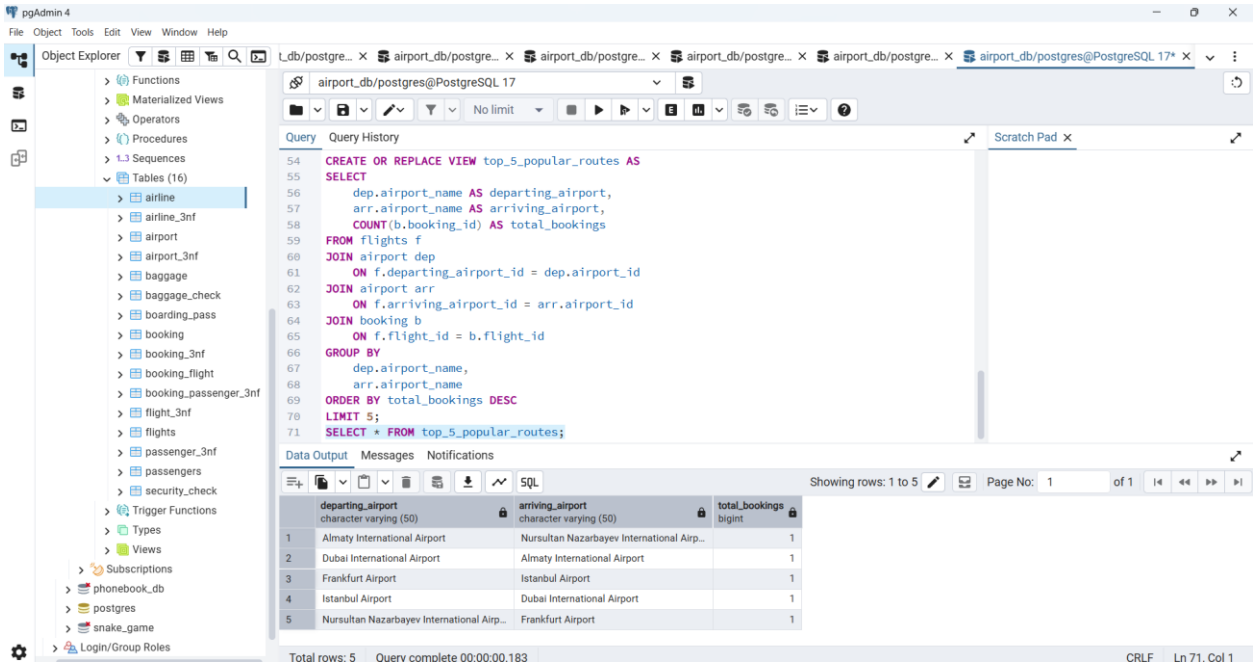
2


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
--2
CREATE OR REPLACE VIEW bookings_next_week AS
SELECT
b.booking_id,
b.passenger_id,
b.booking_platform,
b.status AS booking_status,
b.ticket_price,
f.flight_id,
a.airline_name,
dep.airport_name AS departing_airport,
arr.airport_name AS arriving_airport,
f.sch_departure_time,
f.sch_arrival_time
FROM booking b
JOIN flights f
ON b.flight_id = f.flight_id
JOIN airline a
Data Output Messages Notifications
CREATE VIEW
Query returned successfully in 166 msec.
Total rows: Query complete 00:00:00.166
CRLF Ln 26, Col 1
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (16)
airline
airline_3nf
airport
airport_3nf
baggage
baggage_check
boarding_pass
booking
booking_3nf
booking_flight
booking_passenger_3nf
flight_3nf
flights
passenger_3nf
passengers
security_check
Trigger Functions
Types
Views
Subscriptions
phonebook_db
postgres
snake_game
Login/Group Roles

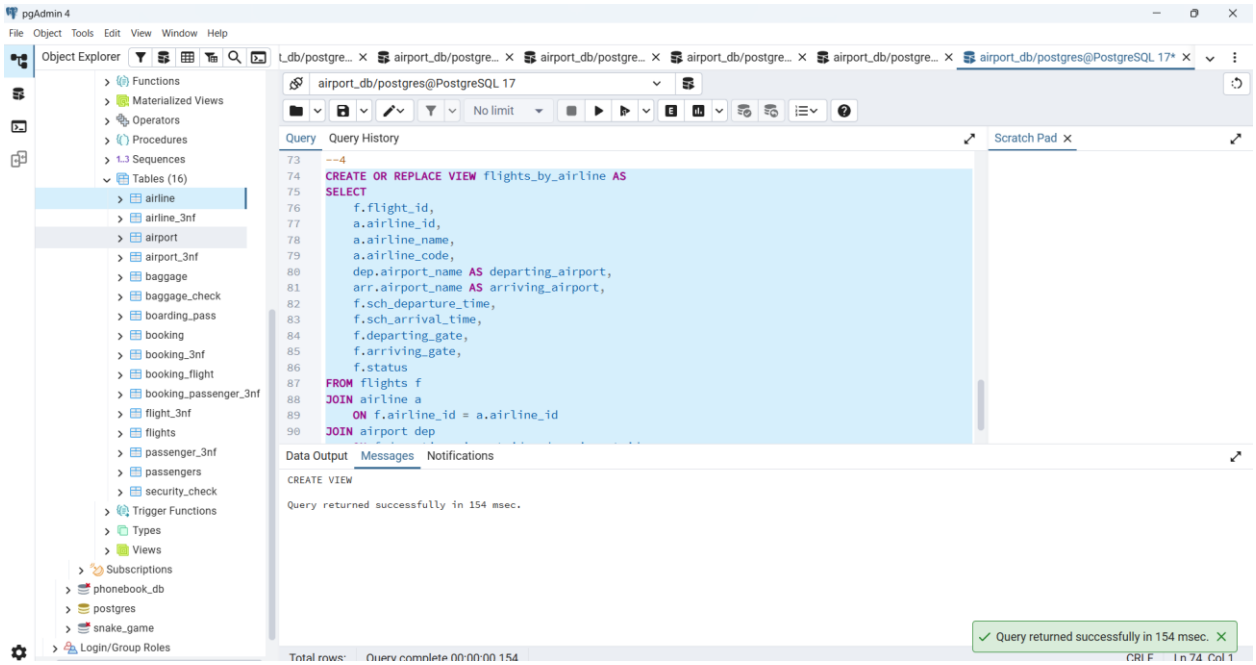
pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
a.airline_name,
dep.airport_name AS departing_airport,
arr.airport_name AS arriving_airport,
f.sch_departure_time,
f.sch_arrival_time
FROM booking b
JOIN flights f
ON b.flight_id = f.flight_id
JOIN airline a
ON f.airline_id = a.airline_id
JOIN airport dep
ON f.departing_airport_id = dep.airport_id
JOIN airport arr
ON f.arriving_airport_id = arr.airport_id
WHERE f.sch_departure_time >= CURRENT_DATE
AND f.sch_departure_time < CURRENT_DATE + INTERVAL '7 days';
SELECT * FROM bookings_next_week;
Data Output Messages Notifications
booking_id integer
passenger_id integer
booking_platform character varying (50)
booking_status character varying (50)
ticket_price numeric (7,2)
flight_id integer
airline_name character varying (50)
departing_airport character varying (50)
arriving_airport character varying (50)
Total rows: 0 Query complete 00:00:00.169
CRLF Ln 51, Col 1

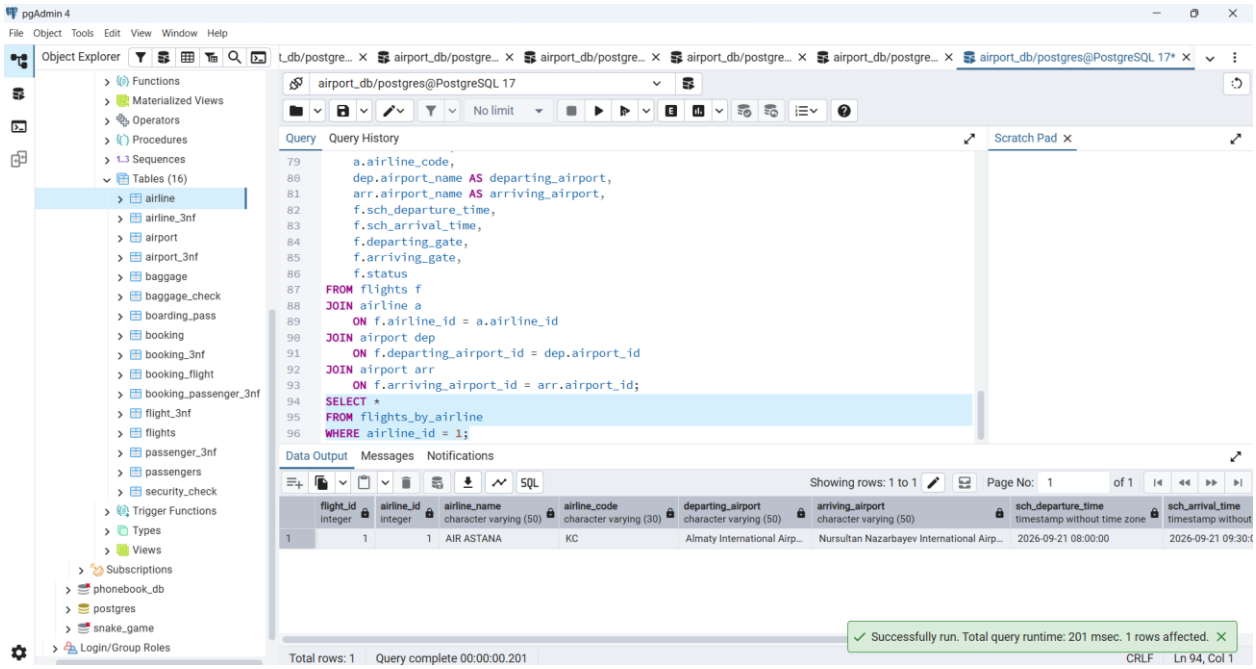
3


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17*
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (16)
airline
airline_3nf
airport
airport_3nf
baggage
baggage_check
boarding_pass
booking
booking_3nf
booking_flight
booking_passenger_3nf
flight_3nf
flights
passenger_3nf
passengers
security_check
Trigger Functions
Types
Views
Subscriptions
phonebook_db
postgres
snake_game
Login/Group Roles
airport_db/postgres@PostgreSQL 17
No limit
Query
Query History
Scratch Pad
53 --3
54 CREATE OR REPLACE VIEW top_5_popular_routes AS
55 SELECT
56 dep.airport_name AS departing_airport,
57 arr.airport_name AS arriving_airport,
58 COUNT(b.booking_id) AS total_bookings
59 FROM flights f
60 JOIN airport dep
61 ON f.departing_airport_id = dep.airport_id
62 JOIN airport arr
63 ON f.arriving_airport_id = arr.airport_id
64 JOIN booking b
65 ON f.flight_id = b.flight_id
66 GROUP BY
67 dep.airport_name,
68 arr.airport_name
69 ORDER BY total_bookings DESC
70 LIMIT 5;
Data Output
Messages
Notifications
CREATE VIEW
Query returned successfully in 167 msec.
Query returned successfully in 167 msec.

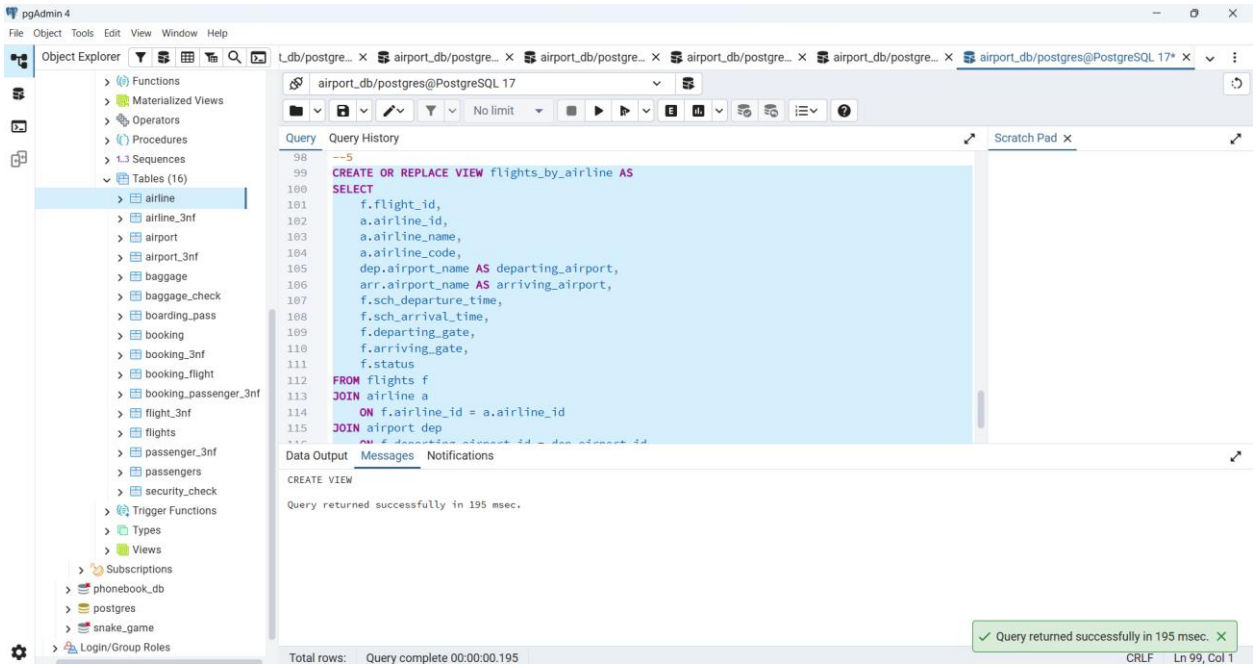

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17*
airport_db/postgres@PostgreSQL 17
No limit
Query
Query History
Scratch Pad
54 CREATE OR REPLACE VIEW top_5_popular_routes AS
55 SELECT
56 dep.airport_name AS departing_airport,
57 arr.airport_name AS arriving_airport,
58 COUNT(b.booking_id) AS total_bookings
59 FROM flights f
60 JOIN airport dep
61 ON f.departing_airport_id = dep.airport_id
62 JOIN airport arr
63 ON f.arriving_airport_id = arr.airport_id
64 JOIN booking b
65 ON f.flight_id = b.flight_id
66 GROUP BY
67 dep.airport_name,
68 arr.airport_name
69 ORDER BY total_bookings DESC
70 LIMIT 5;
71 SELECT * FROM top_5_popular_routes;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 5
Page No: 1 of 1
departing_airport character varying (50)
arriving_airport character varying (50)
total_bookings bigint
1 Almaty International Airport Nursultan Nazarbayev International Airp... 1
2 Dubai International Airport Almaty International Airport 1
3 Frankfurt Airport Istanbul Airport 1
4 Istanbul Airport Dubai International Airport 1
5 Nursultan Nazarbayev International Airp... Frankfurt Airport 1
Total rows: 5
Query complete 00:00:00.183
CRLF
Ln 71, Col 1

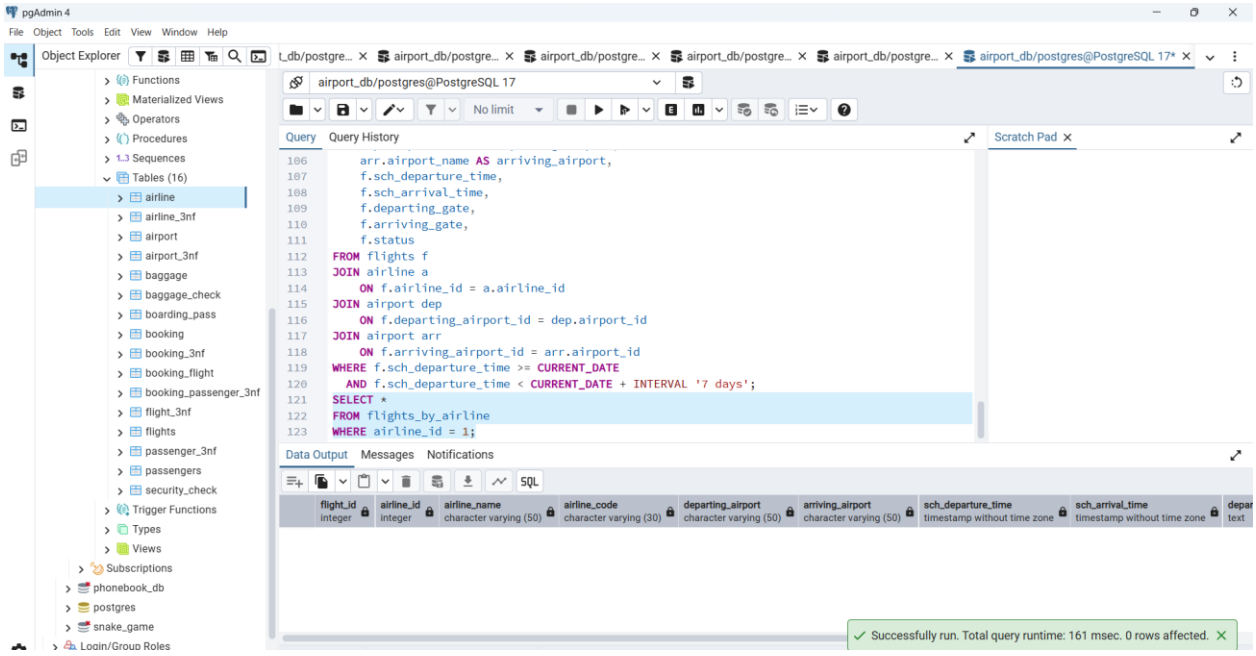
4


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
--4
CREATE OR REPLACE VIEW flights_by_airline AS
SELECT
f.flight_id,
a.airline_id,
a.airline_name,
a.airline_code,
dep.airport_name AS departing_airport,
arr.airport_name AS arriving_airport,
f.sch_departure_time,
f.sch_arrival_time,
f.departing_gate,
f.arriving_gate,
f.status
FROM flights f
JOIN airline a
ON f.airline_id = a.airline_id
JOIN airport dep
Data Output Messages Notifications
CREATE VIEW
Query returned successfully in 154 msec.
Query returned successfully in 154 msec.

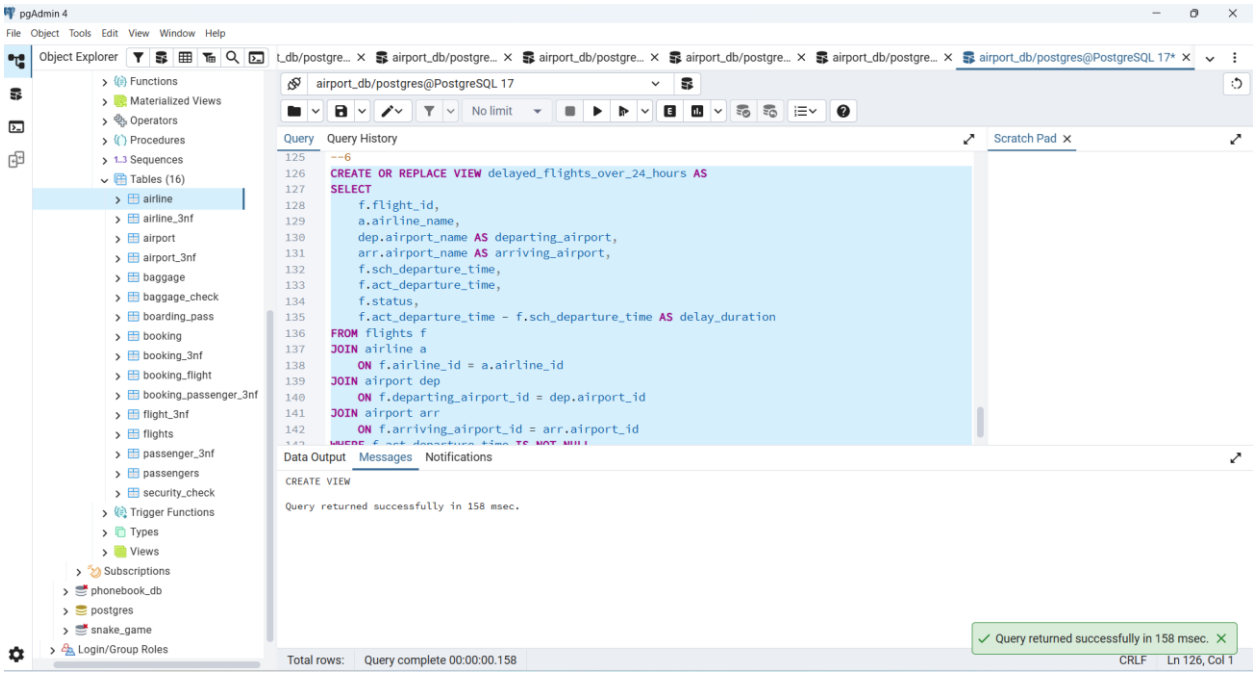

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
a.airline_code,
dep.airport_name AS departing_airport,
arr.airport_name AS arriving_airport,
f.sch_departure_time,
f.sch_arrival_time,
f.departing_gate,
f.arriving_gate,
f.status
FROM flights f
JOIN airline a
ON f.airline_id = a.airline_id
JOIN airport dep
ON f.departing_airport_id = dep.airport_id
JOIN airport arr
ON f.arriving_airport_id = arr.airport_id;
SELECT *
FROM flights_by_airline
WHERE airline_id = 1;
Data Output Messages Notifications
flight_id integer
airline_id integer
airline_name character varying (50)
airline_code character varying (30)
departing_airport character varying (50)
arriving_airport character varying (50)
sch_departure_time timestamp without time zone
sch_arrival_time timestamp without
1 1 1 AIR ASTANA KC Almaty International Airp... Nursultan Nazarbayev International Airp... 2026-09-21 08:00:00 2026-09-21 09:30:0
Successfully run. Total query runtime: 201 msec. 1 rows affected.
Total rows: 1 Query complete 00:00:00.201

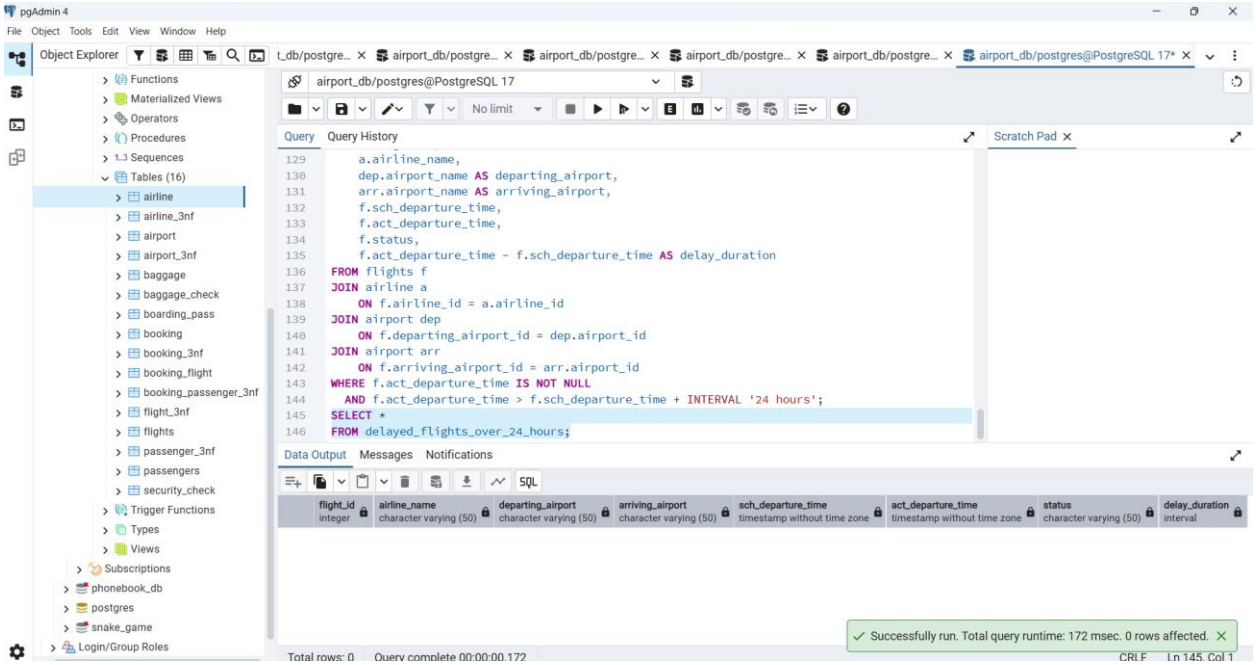
5


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
--5
CREATE OR REPLACE VIEW flights_by_airline AS
SELECT
f.flight_id,
a.airline_id,
a.airline_name,
a.airline_code,
dep.airport_name AS departing_airport,
arr.airport_name AS arriving_airport,
f.sch_departure_time,
f.sch_arrival_time,
f.departing_gate,
f.arriving_gate,
f.status
FROM flights f
JOIN airline a
ON f.airline_id = a.airline_id
JOIN airport dep
Data Output Messages Notifications
CREATE VIEW
Query returned successfully in 195 msec.
Query returned successfully in 195 msec.
Total rows:
Query complete 00:00:00.195
CRLF
Ln 99, Col 1

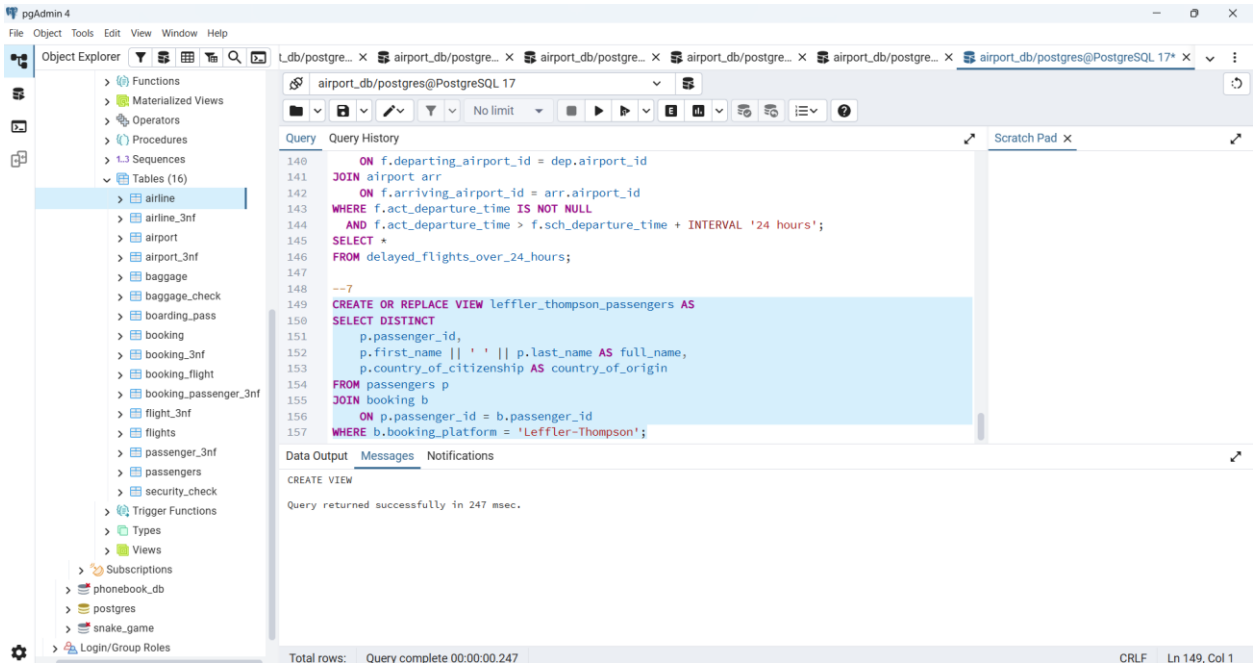

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
arr.airport_name AS arriving_airport,
f.sch_departure_time,
f.sch_arrival_time,
f.departing_gate,
f.arriving_gate,
f.status
FROM flights f
JOIN airline a
ON f.airline_id = a.airline_id
JOIN airport dep
ON f.departing_airport_id = dep.airport_id
JOIN airport arr
ON f.arriving_airport_id = arr.airport_id
WHERE f.sch_departure_time >= CURRENT_DATE
AND f.sch_departure_time < CURRENT_DATE + INTERVAL '7 days';
SELECT *
FROM flights_by_airline
WHERE airline_id = 1;
Data Output Messages Notifications
flight_id integer
airline_id integer
airline_name character varying (50)
airline_code character varying (30)
departing_airport character varying (50)
arriving_airport character varying (50)
sch_departure_time timestamp without time zone
sch_arrival_time timestamp without time zone
Successfully run. Total query runtime: 161 msec. 0 rows affected.

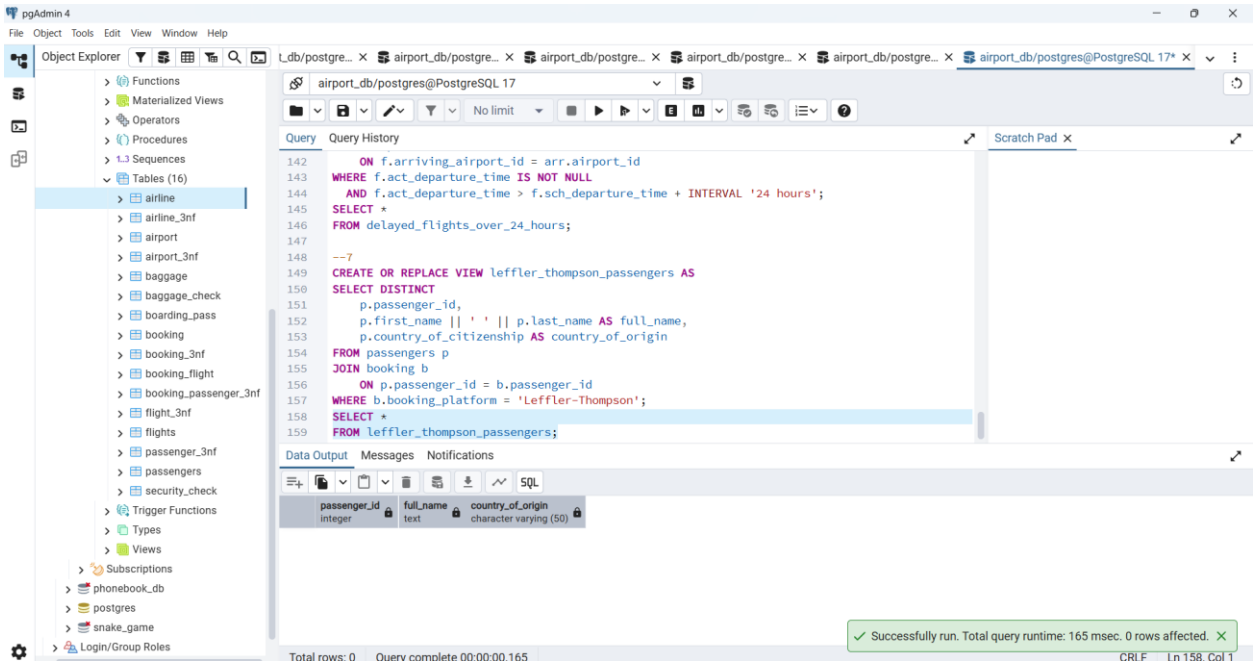
6


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17*
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (16)
airline
airline_3nf
airport
airport_3nf
baggage
baggage_check
boarding_pass
booking
booking_3nf
booking_flight
booking_passenger_3nf
flight_3nf
flights
passenger_3nf
passengers
security_check
Trigger Functions
Types
Views
Subscriptions
phonebook_db
postgres
snake_game
Login/Group Roles
Query
Query History
Scratch Pad
--6
CREATE OR REPLACE VIEW delayed_flights_over_24_hours AS
SELECT
f.flight_id,
a.airline_name,
dep.airport_name AS departing_airport,
arr.airport_name AS arriving_airport,
f.sch_departure_time,
f.act_departure_time,
f.status,
f.act_departure_time - f.sch_departure_time AS delay_duration
FROM flights f
JOIN airline a
ON f.airline_id = a.airline_id
JOIN airport dep
ON f.departing_airport_id = dep.airport_id
JOIN airport arr
ON f.arriving_airport_id = arr.airport_id
Data Output
Messages
Notifications
CREATE VIEW
Query returned successfully in 158 msec.
Total rows:
Query complete 00:00:00.158
CRLF
Ln 126, Col 1

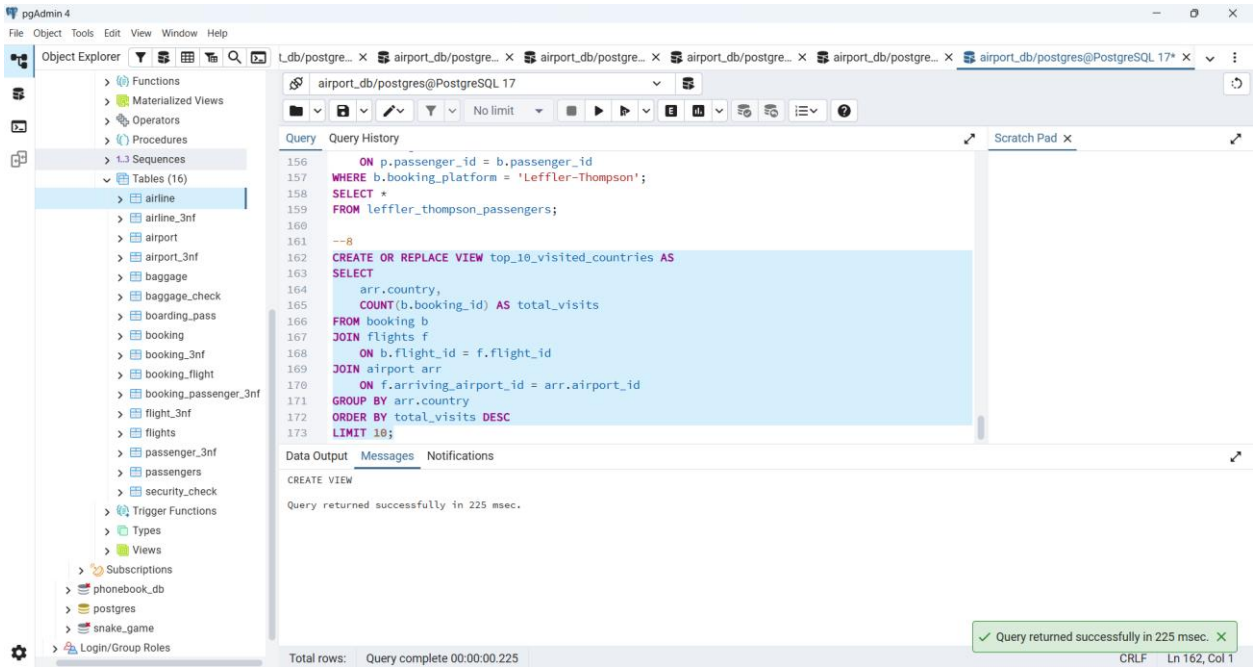

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17*
Query
Query History
Scratch Pad
a.airline_name,
dep.airport_name AS departing_airport,
arr.airport_name AS arriving_airport,
f.sch_departure_time,
f.act_departure_time,
f.status,
f.act_departure_time - f.sch_departure_time AS delay_duration
FROM flights f
JOIN airline a
ON f.airline_id = a.airline_id
JOIN airport dep
ON f.departing_airport_id = dep.airport_id
JOIN airport arr
ON f.arriving_airport_id = arr.airport_id
WHERE f.act_departure_time IS NOT NULL
AND f.act_departure_time > f.sch_departure_time + INTERVAL '24 hours';
SELECT *
FROM delayed_flights_over_24_hours;
Data Output
Messages
Notifications
flight_id integer
airline_name character varying (50)
departing_airport character varying (50)
arriving_airport character varying (50)
sch_departure_time timestamp without time zone
act_departure_time timestamp without time zone
status character varying (50)
delay_duration interval
Successfully run. Total query runtime: 172 msec. 0 rows affected.
Total rows: 0
Query complete 00:00:00.172
CRLF
Ln 145, Col 1

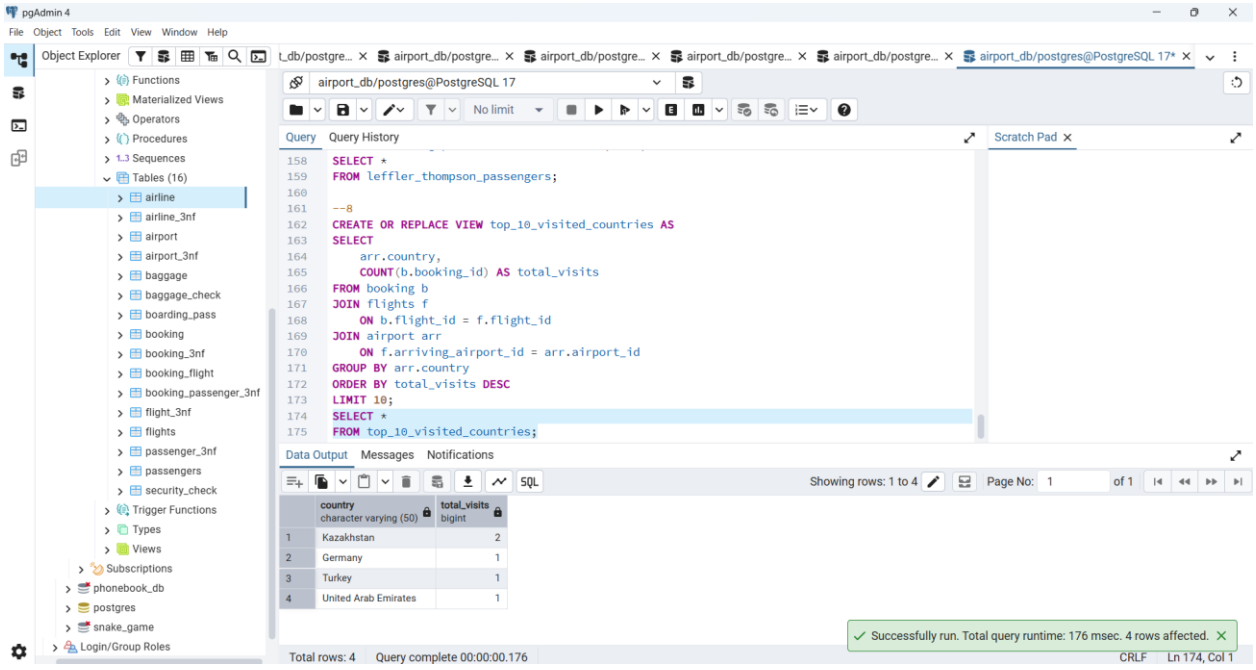
7


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
No limit
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (16)
airline
airline_3nf
airport
airport_3nf
baggage
baggage_check
boarding_pass
booking
booking_3nf
booking_flight
booking_passenger_3nf
flight_3nf
flights
passenger_3nf
passengers
security_check
Trigger Functions
Types
Views
Subscriptions
phonebook_db
postgres
snake_game
Login/Group Roles
140 ON f.departing_airport_id = dep.airport_id
141 JOIN airport arr
142 ON f.arriving_airport_id = arr.airport_id
143 WHERE f.act_departure_time IS NOT NULL
144 AND f.act_departure_time > f.sch_departure_time + INTERVAL '24 hours';
145 SELECT *
146 FROM delayed_flights_over_24_hours;
147
148 --7
149 CREATE OR REPLACE VIEW leffler_thompson_passengers AS
150 SELECT DISTINCT
151 p.passenger_id,
152 p.first_name || ' ' || p.last_name AS full_name,
153 p.country_of_citizenship AS country_of_origin
154 FROM passengers p
155 JOIN booking b
156 ON p.passenger_id = b.passenger_id
157 WHERE b.booking_platform = 'Leffler-Thompson';
Data Output Messages Notifications
CREATE VIEW
Query returned successfully in 247 msec.
Total rows:
Query complete 00:00:00.247
CRLF
Ln 149, Col 1

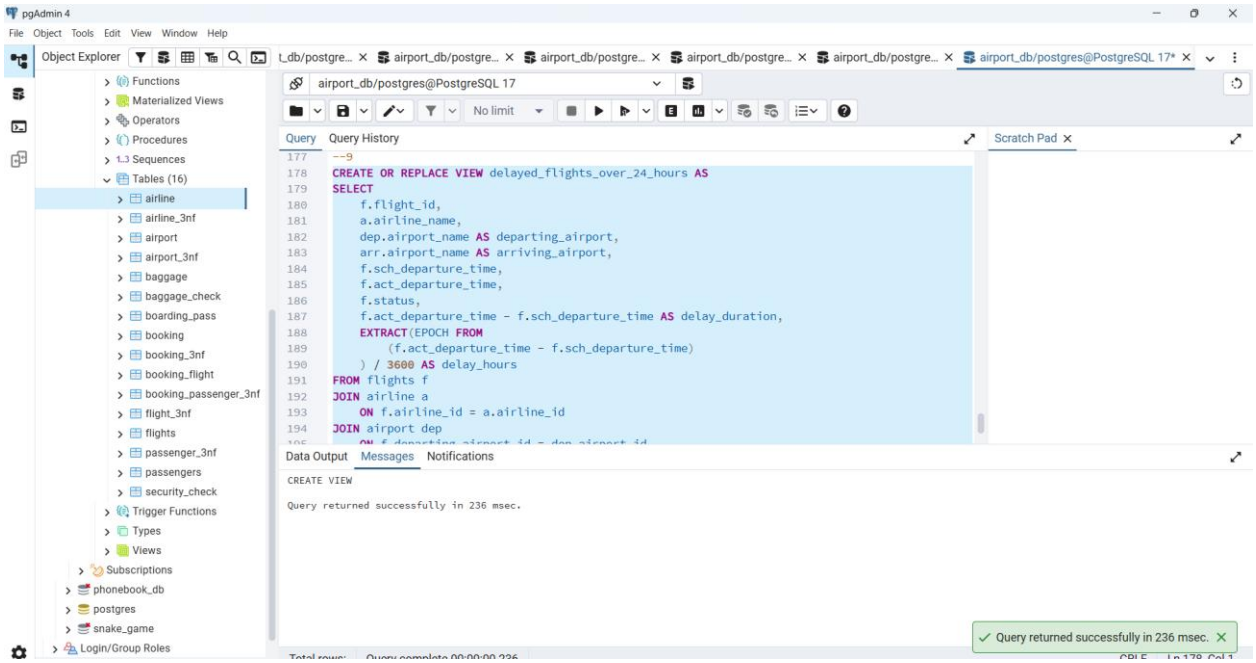

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
No limit
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (16)
airline
airline_3nf
airport
airport_3nf
baggage
baggage_check
boarding_pass
booking
booking_3nf
booking_flight
booking_passenger_3nf
flight_3nf
flights
passenger_3nf
passengers
security_check
Trigger Functions
Types
Views
Subscriptions
phonebook_db
postgres
snake_game
Login/Group Roles
142 ON f.arriving_airport_id = arr.airport_id
143 WHERE f.act_departure_time IS NOT NULL
144 AND f.act_departure_time > f.sch_departure_time + INTERVAL '24 hours';
145 SELECT *
146 FROM delayed_flights_over_24_hours;
147
148 --7
149 CREATE OR REPLACE VIEW leffler_thompson_passengers AS
150 SELECT DISTINCT
151 p.passenger_id,
152 p.first_name || ' ' || p.last_name AS full_name,
153 p.country_of_citizenship AS country_of_origin
154 FROM passengers p
155 JOIN booking b
156 ON p.passenger_id = b.passenger_id
157 WHERE b.booking_platform = 'Leffler-Thompson';
158 SELECT *
159 FROM leffler_thompson_passengers;
Data Output Messages Notifications
SQL
passenger_id integer
full_name text
country_of_origin character varying (50)
Successfully run. Total query runtime: 165 msec. 0 rows affected.
Total rows: 0
Query complete 00:00:00.165
CRLF
Ln 158, Col 1

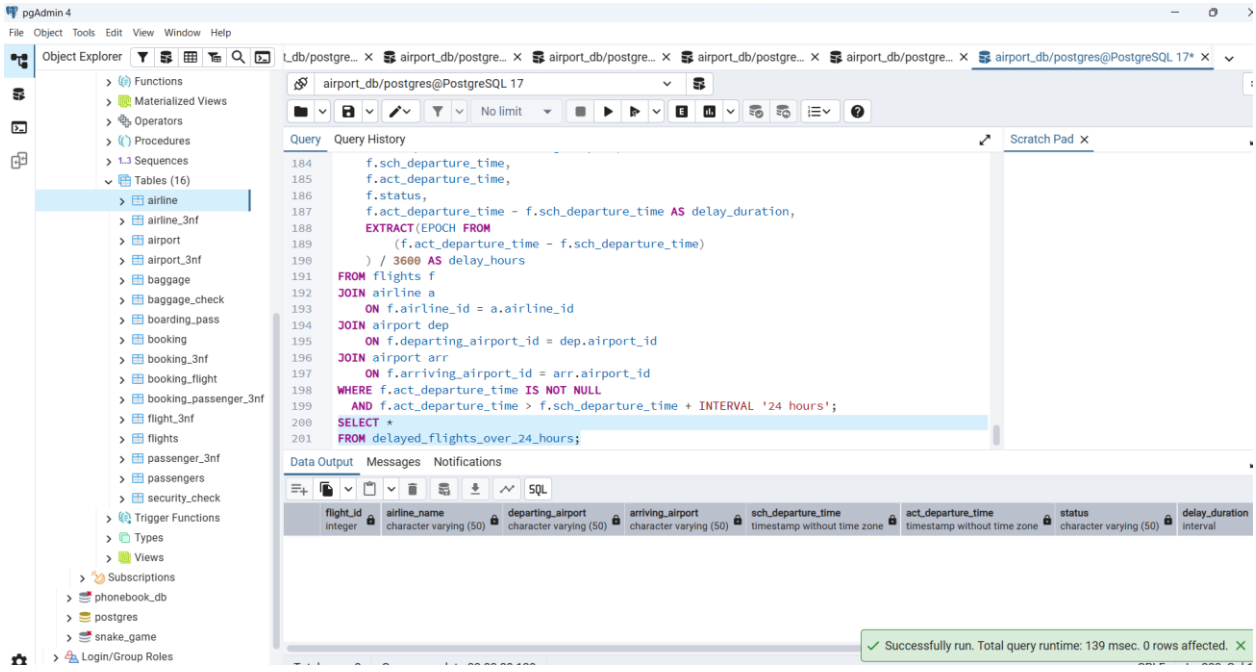
8


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (16)
airline
airline_3nf
airport
airport_3nf
baggage
baggage_check
boarding_pass
booking
booking_3nf
booking_flight
booking_passenger_3nf
flight_3nf
flights
passenger_3nf
passengers
security_check
Trigger Functions
Types
Views
Subscriptions
phonebook_db
postgres
snake_game
Login/Group Roles
156 ON p.passenger_id = b.passenger_id
157 WHERE b.booking_platform = 'Leffler-Thompson';
158 SELECT *
159 FROM leffler_thompson_passengers;
160
161 --8
162 CREATE OR REPLACE VIEW top_10_visited_countries AS
163 SELECT
164 arr.country,
165 COUNT(b.booking_id) AS total_visits
166 FROM booking b
167 JOIN flights f
168 ON b.flight_id = f.flight_id
169 JOIN airport arr
170 ON f.arriving_airport_id = arr.airport_id
171 GROUP BY arr.country
172 ORDER BY total_visits DESC
173 LIMIT 10;
Data Output Messages Notifications
CREATE VIEW
Query returned successfully in 225 msec.
Query returned successfully in 225 msec.
Total rows:
Query complete 00:00:00.225
CRLF
Ln 162, Col 1

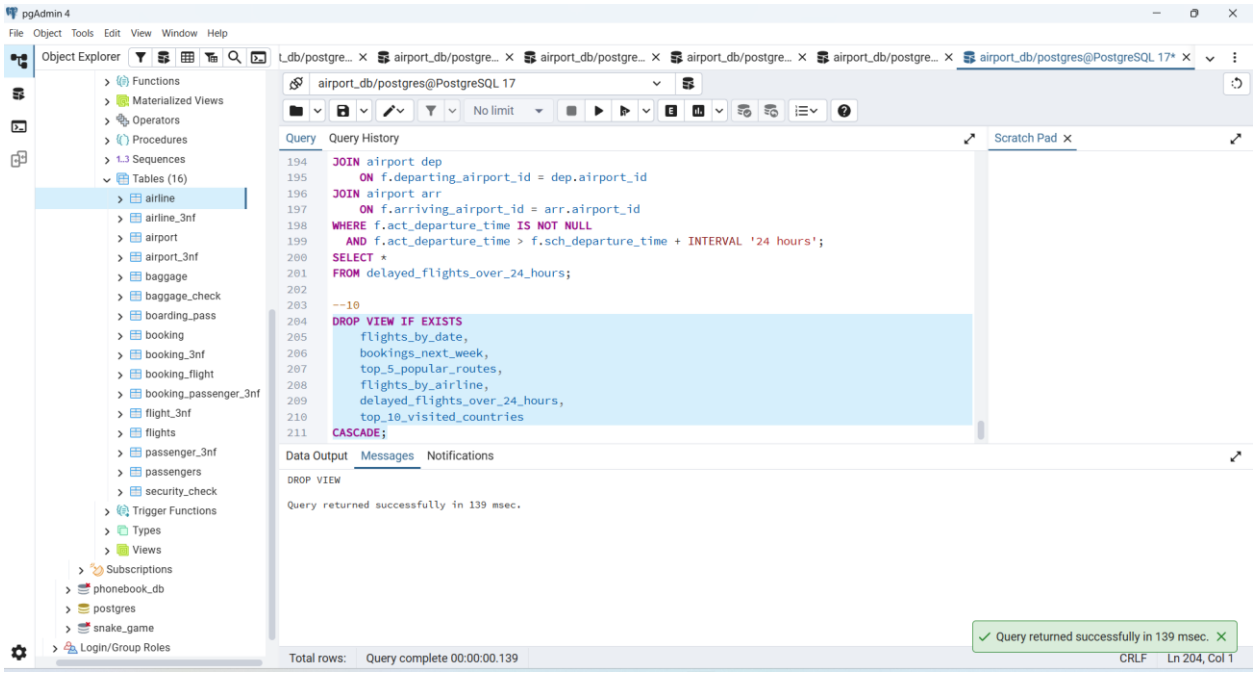

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
158 SELECT *
159 FROM leffler_thompson_passengers;
160
161 --8
162 CREATE OR REPLACE VIEW top_10_visited_countries AS
163 SELECT
164 arr.country,
165 COUNT(b.booking_id) AS total_visits
166 FROM booking b
167 JOIN flights f
168 ON b.flight_id = f.flight_id
169 JOIN airport arr
170 ON f.arriving_airport_id = arr.airport_id
171 GROUP BY arr.country
172 ORDER BY total_visits DESC
173 LIMIT 10;
174 SELECT *
175 FROM top_10_visited_countries;
Data Output Messages Notifications
Showing rows: 1 to 4
Page No: 1 of 1
country character varying (50)
total_visits bigint
1 Kazakhstan 2
2 Germany 1
3 Turkey 1
4 United Arab Emirates 1
Successfully run. Total query runtime: 176 msec. 4 rows affected.
Total rows: 4
Query complete 00:00:00.176
CRLF
Ln 174, Col 1

9

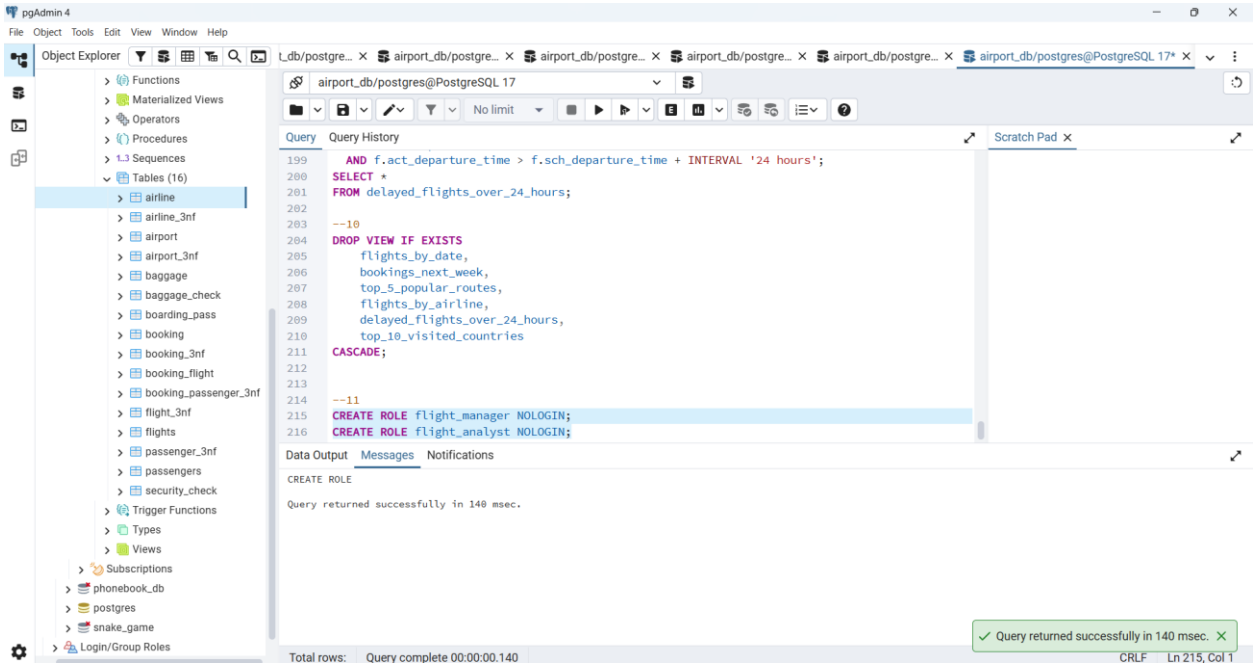

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
--9
CREATE OR REPLACE VIEW delayed_flights_over_24_hours AS
SELECT
f.flight_id,
a.airline_name,
dep.airport_name AS departing_airport,
arr.airport_name AS arriving_airport,
f.sch_departure_time,
f.act_departure_time,
f.status,
f.act_departure_time - f.sch_departure_time AS delay_duration,
EXTRACT(EPOCH FROM
(f.act_departure_time - f.sch_departure_time)
) / 3600 AS delay_hours
FROM flights f
JOIN airline a
ON f.airline_id = a.airline_id
JOIN airport dep
Data Output Messages Notifications
CREATE VIEW
Query returned successfully in 236 msec.
Query returned successfully in 236 msec.

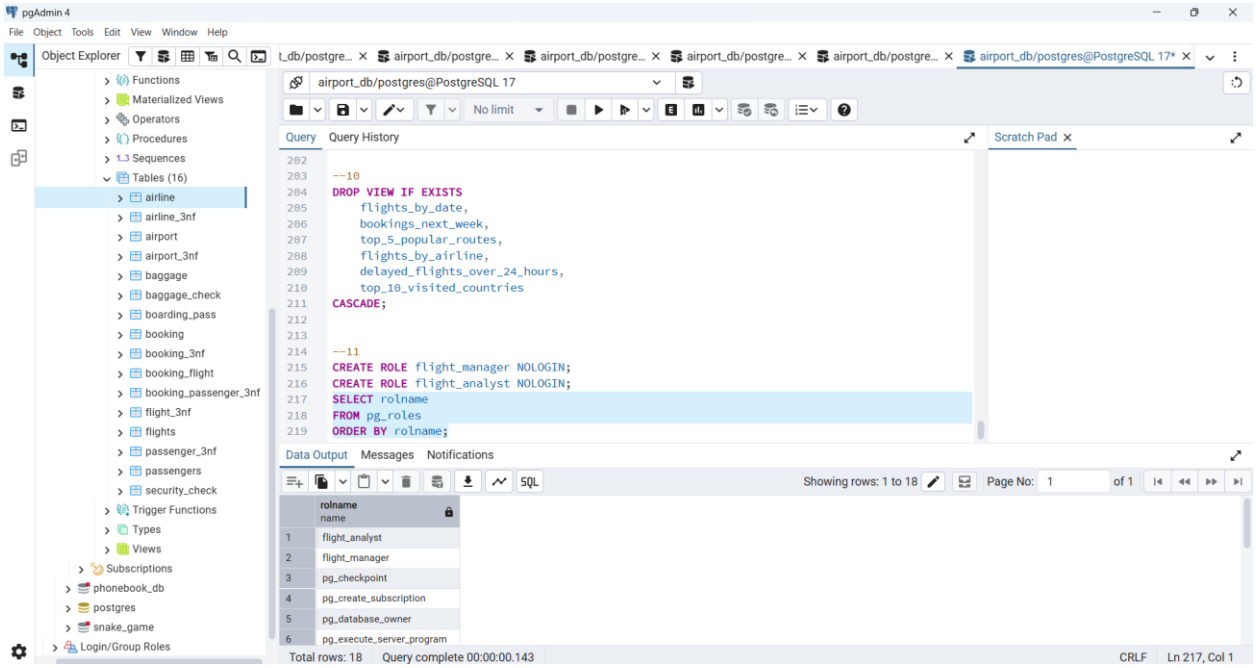

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
f.sch_departure_time,
f.act_departure_time,
f.status,
f.act_departure_time - f.sch_departure_time AS delay_duration,
EXTRACT(EPOCH FROM
(f.act_departure_time - f.sch_departure_time)
) / 3600 AS delay_hours
FROM flights f
JOIN airline a
ON f.airline_id = a.airline_id
JOIN airport dep
ON f.departing_airport_id = dep.airport_id
JOIN airport arr
ON f.arriving_airport_id = arr.airport_id
WHERE f.act_departure_time IS NOT NULL
AND f.act_departure_time > f.sch_departure_time + INTERVAL '24 hours';
SELECT *
FROM delayed_flights_over_24_hours;
Data Output Messages Notifications
flight_id integer
airline_name character varying (50)
departing_airport character varying (50)
arriving_airport character varying (50)
sch_departure_time timestamp without time zone
act_departure_time timestamp without time zone
status character varying (50)
delay_duration interval
Successfully run. Total query runtime: 139 msec. 0 rows affected.

10

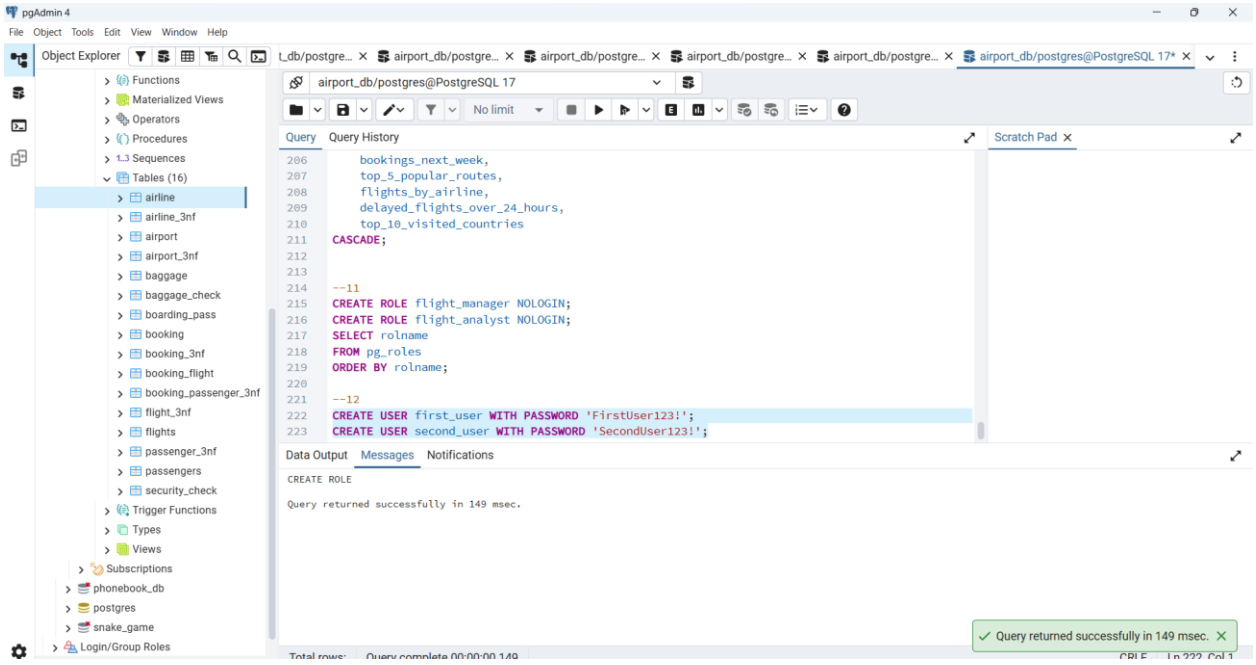

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
JOIN airport dep
ON f.departing_airport_id = dep.airport_id
JOIN airport arr
ON f.arriving_airport_id = arr.airport_id
WHERE f.act_departure_time IS NOT NULL
AND f.act_departure_time > f.sch_departure_time + INTERVAL '24 hours';
SELECT *
FROM delayed_flights_over_24_hours;
--10
DROP VIEW IF EXISTS
flights_by_date,
bookings_next_week,
top_5_popular_routes,
flights_by_airline,
delayed_flights_over_24_hours,
top_10_visited_countries
CASCADE;
Data Output Messages Notifications
DROP VIEW
Query returned successfully in 139 msec.
Query returned successfully in 139 msec.
Total rows: Query complete 00:00:00.139
CRLF Ln 204, Col 1

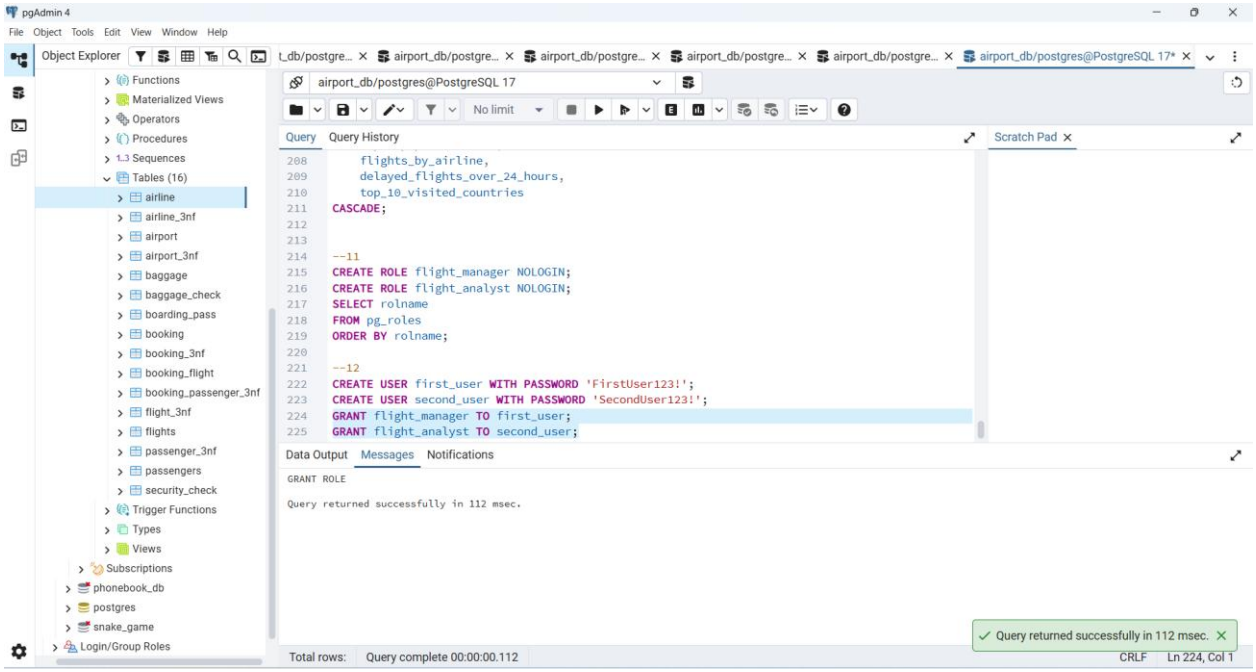
11

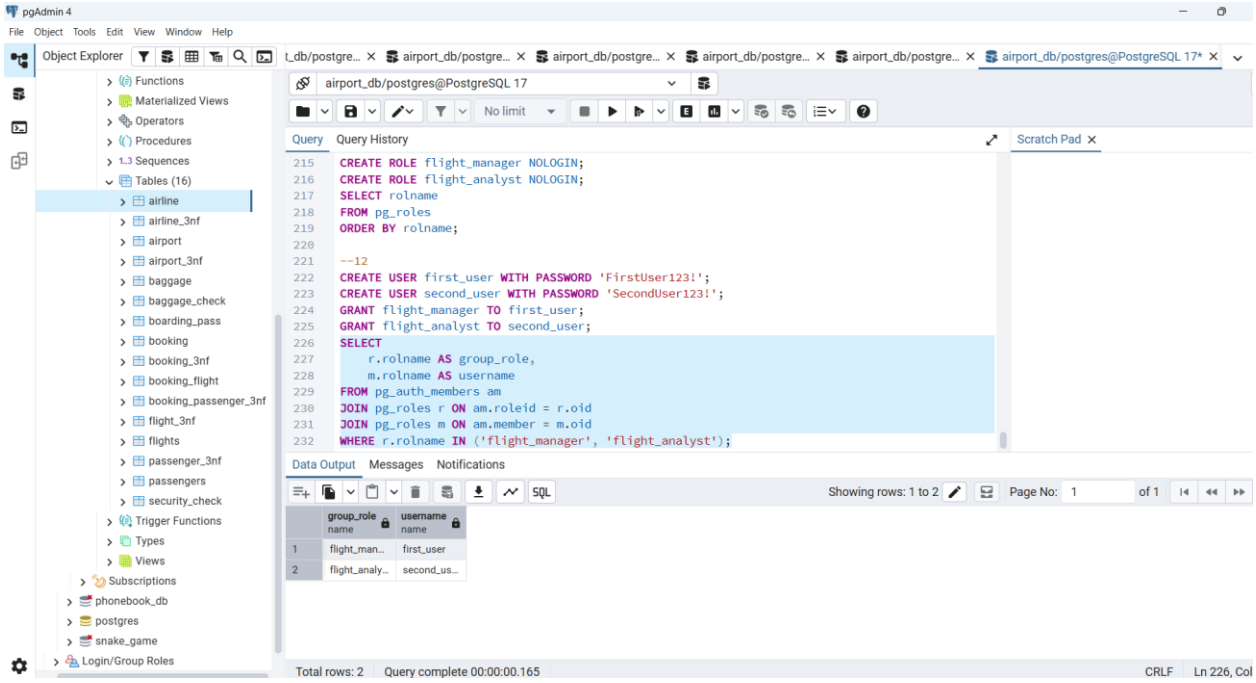

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
AND f.act_departure_time > f.sch_departure_time + INTERVAL '24 hours';
SELECT *
FROM delayed_flights_over_24_hours;
--10
DROP VIEW IF EXISTS
flights_by_date,
bookings_next_week,
top_5_popular_routes,
flights_by_airline,
delayed_flights_over_24_hours,
top_10_visited_countries
CASCADE;
--11
CREATE ROLE flight_manager NOLOGIN;
CREATE ROLE flight_analyst NOLOGIN;
Data Output Messages Notifications
CREATE ROLE
Query returned successfully in 140 msec.
Query returned successfully in 140 msec.
Total rows: Query complete 00:00:00.140
CRLF Ln 215, Col 1


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
No limit
Query Query History
Scratch Pad
--10
DROP VIEW IF EXISTS
flights_by_date,
bookings_next_week,
top_5_popular_routes,
flights_by_airline,
delayed_flights_over_24_hours,
top_10_visited_countries
CASCADE;
--11
CREATE ROLE flight_manager NOLOGIN;
CREATE ROLE flight_analyst NOLOGIN;
SELECT rolname
FROM pg_roles
ORDER BY rolname;
Data Output Messages Notifications
Showing rows: 1 to 18
Page No: 1 of 1
rolname
name
1 flight_analyst
2 flight_manager
3 pg_checkpoint
4 pg_create_subscription
5 pg_database_owner
6 pg_execute_server_program
Total rows: 18
Query complete 00:00:00.143
CRLF
Ln 217, Col 1

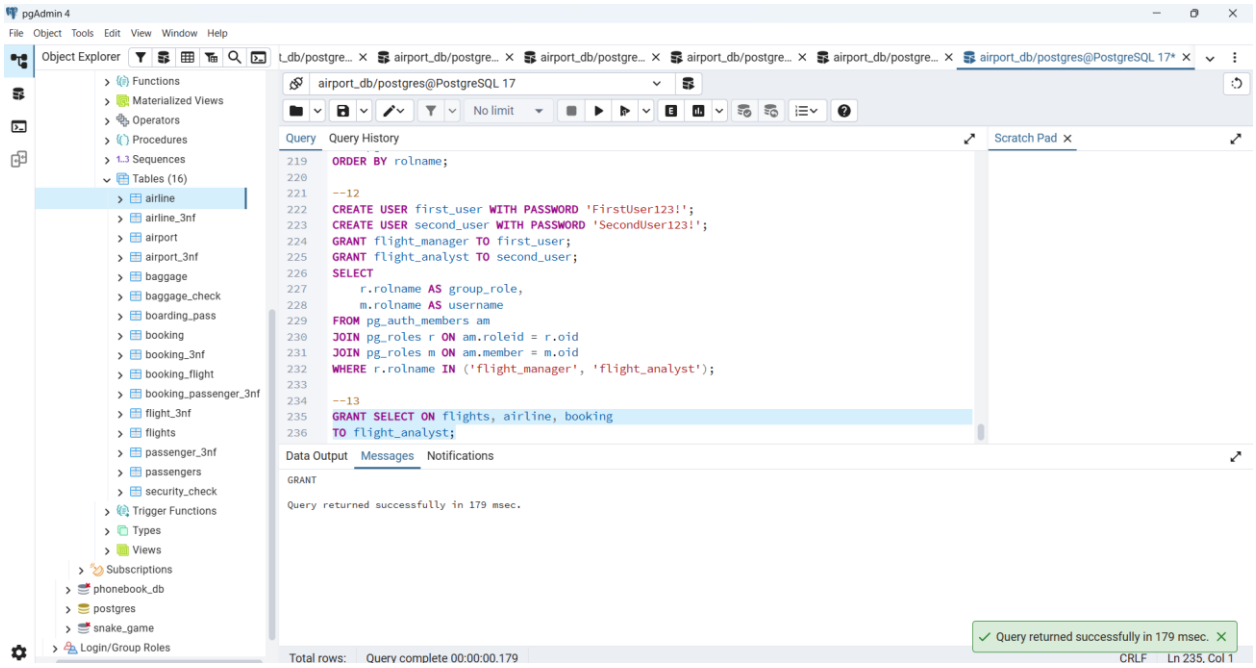
12

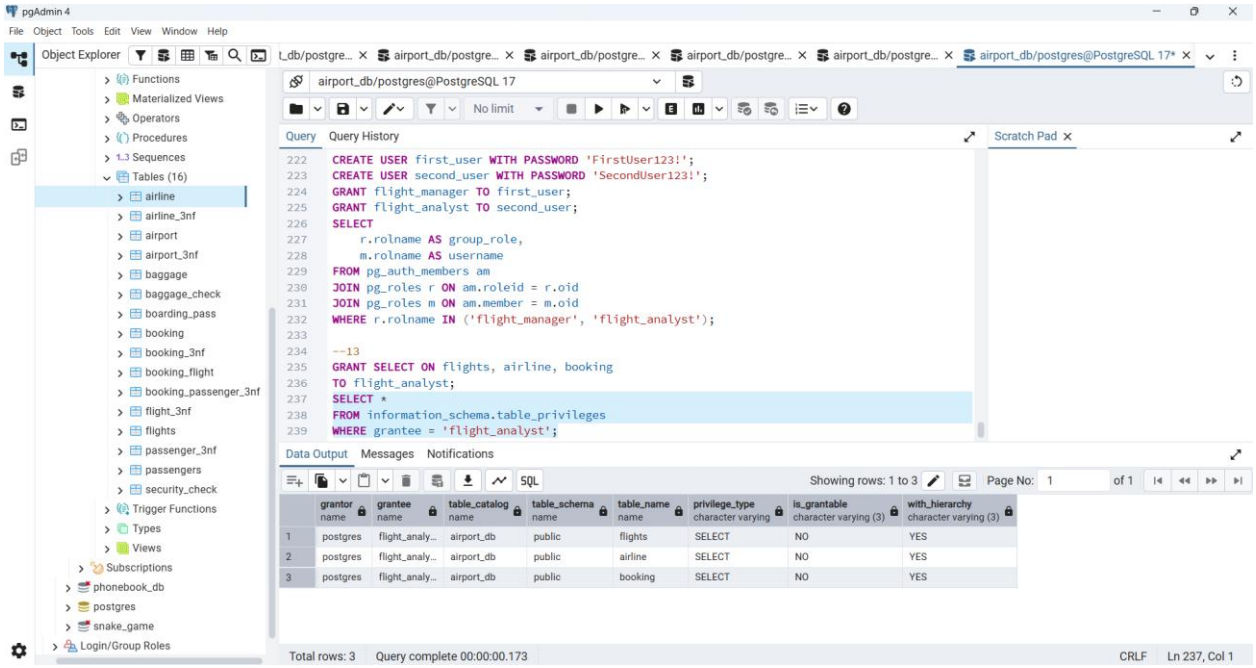

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
No limit
Query Query History
Scratch Pad
bookings_next_week,
top_5_popular_routes,
flights_by_airline,
delayed_flights_over_24_hours,
top_10_visited_countries
CASCADE;
--11
CREATE ROLE flight_manager NOLOGIN;
CREATE ROLE flight_analyst NOLOGIN;
SELECT rolname
FROM pg_roles
ORDER BY rolname;
--12
CREATE USER first_user WITH PASSWORD 'FirstUser123!';
CREATE USER second_user WITH PASSWORD 'SecondUser123!';
Data Output Messages Notifications
CREATE ROLE
Query returned successfully in 149 msec.
Query returned successfully in 149 msec.


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17*
airport_db/postgres@PostgreSQL 17
No limit
Query Query History
Scratch Pad
208 flights_by_airline,
209 delayed_flights_over_24_hours,
210 top_10_visited_countries
211 CASCADE;
214 --11
215 CREATE ROLE flight_manager NOLOGIN;
216 CREATE ROLE flight_analyst NOLOGIN;
217 SELECT rolname
218 FROM pg_roles
219 ORDER BY rolname;
221 --12
222 CREATE USER first_user WITH PASSWORD 'FirstUser123!';
223 CREATE USER second_user WITH PASSWORD 'SecondUser123!';
224 GRANT flight_manager TO first_user;
225 GRANT flight_analyst TO second_user;
Data Output Messages Notifications
GRANT ROLE
Query returned successfully in 112 msec.
Query returned successfully in 112 msec.
Total rows: Query complete 00:00:00.112
CRLF Ln 224, Col 1
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (16)
airline
airline_3nf
airport
airport_3nf
baggage
baggage_check
boarding_pass
booking
booking_3nf
booking_flight
booking_passenger_3nf
flight_3nf
flights
passenger_3nf
passengers
security_check
Trigger Functions
Types
Views
Subscriptions
phonebook_db
postgres
snake_game
Login/Group Roles

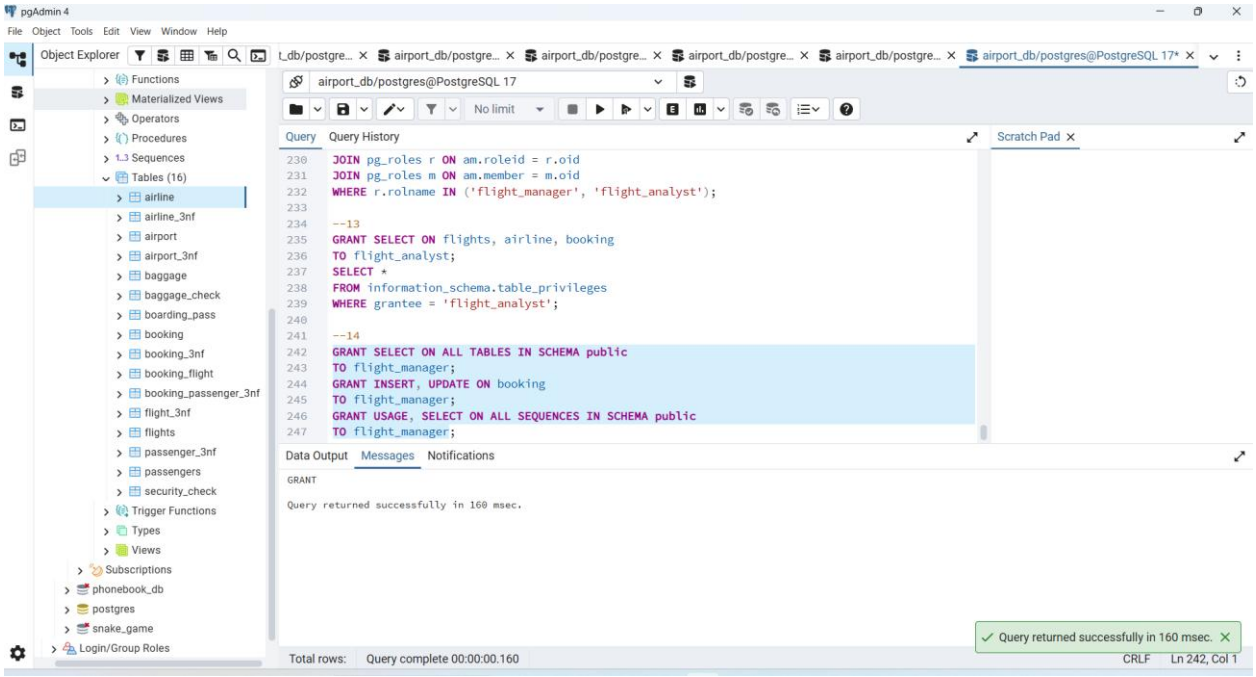

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17*
airport_db/postgres@PostgreSQL 17
No limit
Query Query History
Scratch Pad
215 CREATE ROLE flight_manager NOLOGIN;
216 CREATE ROLE flight_analyst NOLOGIN;
217 SELECT rolname
218 FROM pg_roles
219 ORDER BY rolname;
221 --12
222 CREATE USER first_user WITH PASSWORD 'FirstUser123!';
223 CREATE USER second_user WITH PASSWORD 'SecondUser123!';
224 GRANT flight_manager TO first_user;
225 GRANT flight_analyst TO second_user;
226 SELECT
227 r.rolname AS group_role,
228 m.rolname AS username
229 FROM pg_auth_members am
230 JOIN pg_roles r ON am.roleid = r.oid
231 JOIN pg_roles m ON am.member = m.oid
232 WHERE r.rolname IN ('flight_manager', 'flight_analyst');
Data Output Messages Notifications
SQL
Showing rows: 1 to 2
Page No: 1 of 1
group_role name
username name
1 flight_man... first_user
2 flight_analy... second_us...
Total rows: 2 Query complete 00:00:00.165
CRLF Ln 226, Col
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (16)
airline
airline_3nf
airport
airport_3nf
baggage
baggage_check
boarding_pass
booking
booking_3nf
booking_flight
booking_passenger_3nf
flight_3nf
flights
passenger_3nf
passengers
security_check
Trigger Functions
Types
Views
Subscriptions
phonebook_db
postgres
snake_game
Login/Group Roles

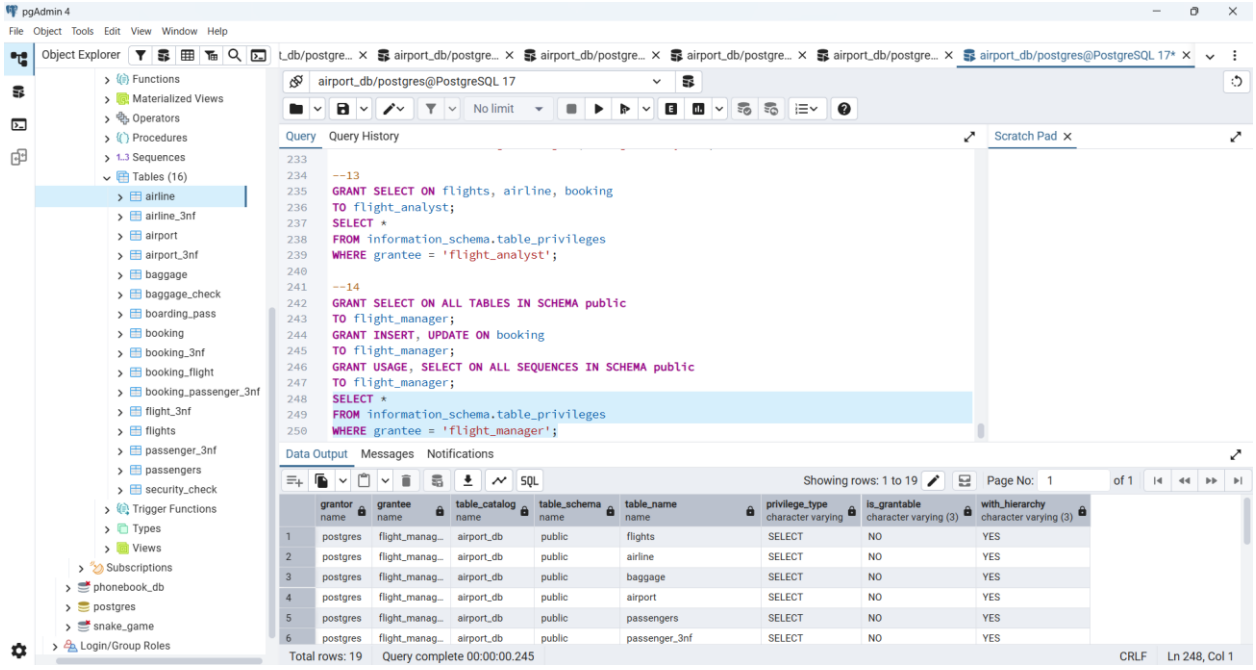
13


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
ORDER BY rolname;
--12
CREATE USER first_user WITH PASSWORD 'FirstUser123!';
CREATE USER second_user WITH PASSWORD 'SecondUser123!';
GRANT flight_manager TO first_user;
GRANT flight_analyst TO second_user;
SELECT
r.rolname AS group_role,
m.rolname AS username
FROM pg_auth_members am
JOIN pg_roles r ON am.roleid = r.oid
JOIN pg_roles m ON am.member = m.oid
WHERE r.rolname IN ('flight_manager', 'flight_analyst');
--13
GRANT SELECT ON flights, airline, booking
TO flight_analyst;
Data Output Messages Notifications
GRANT
Query returned successfully in 179 msec.
Query returned successfully in 179 msec.
Total rows: Query complete 00:00:00.179
CRLF Ln 235, Col 1

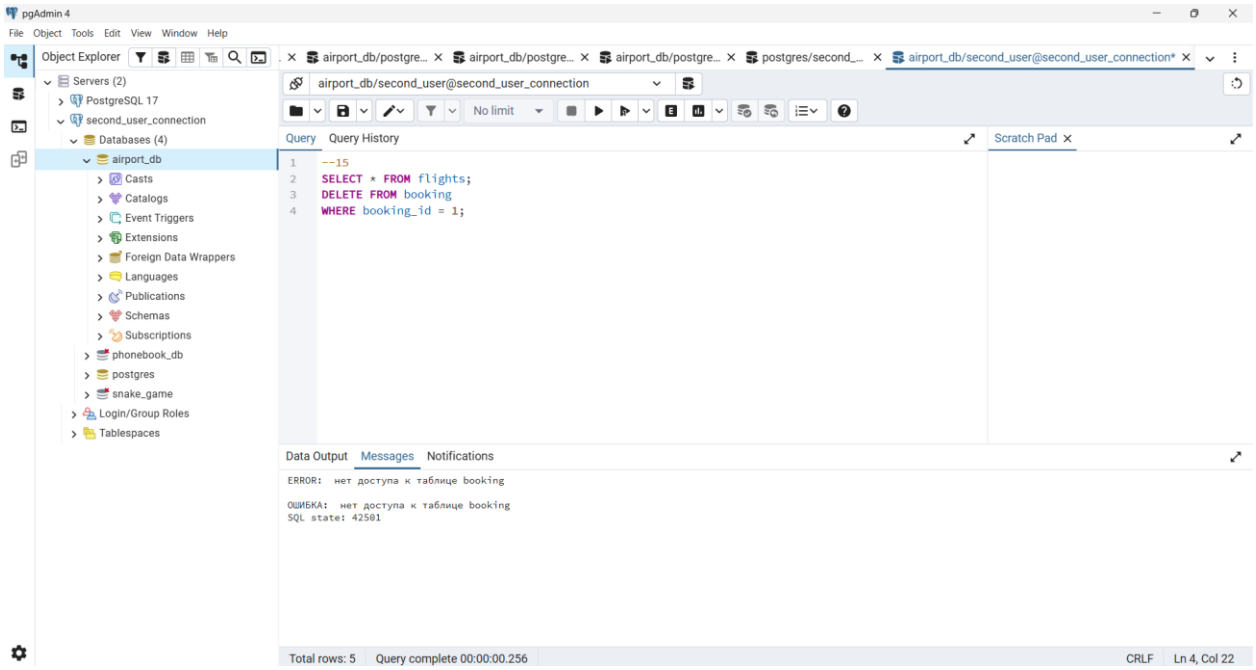

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
CREATE USER first_user WITH PASSWORD 'FirstUser123!';
CREATE USER second_user WITH PASSWORD 'SecondUser123!';
GRANT flight_manager TO first_user;
GRANT flight_analyst TO second_user;
SELECT
r.rolname AS group_role,
m.rolname AS username
FROM pg_auth_members am
JOIN pg_roles r ON am.roleid = r.oid
JOIN pg_roles m ON am.member = m.oid
WHERE r.rolname IN ('flight_manager', 'flight_analyst');
--13
GRANT SELECT ON flights, airline, booking
TO flight_analyst;
SELECT *
FROM information_schema.table_privileges
WHERE grantee = 'flight_analyst';
Data Output Messages Notifications
Showing rows: 1 to 3
Page No: 1 of 1
grantor name
grantee name
table_catalog name
table_schema name
table_name name
privilege_type character varying
is_grantable character varying (3)
with_hierarchy character varying (3)
1 postgres flight_analy... airport_db public flights SELECT NO YES
2 postgres flight_analy... airport_db public airline SELECT NO YES
3 postgres flight_analy... airport_db public booking SELECT NO YES
Total rows: 3 Query complete 00:00:00.173
CRLF Ln 237, Col 1

14

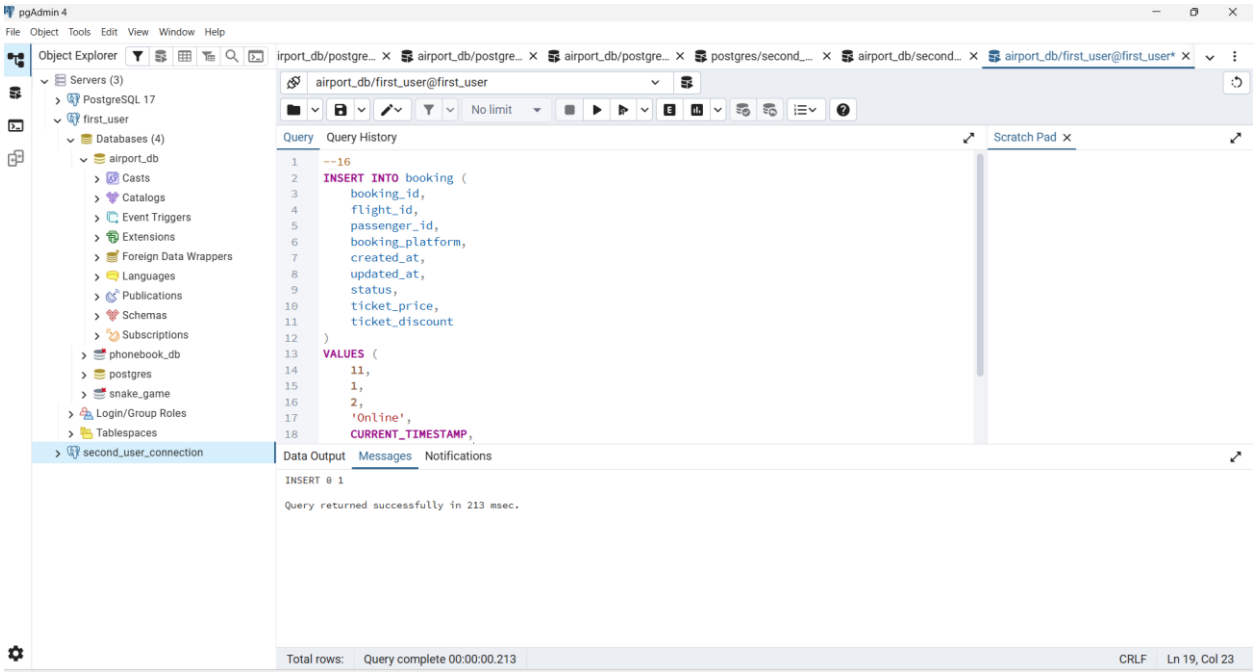

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
JOIN pg_roles r ON am.roleid = r.oid
JOIN pg_roles m ON am.member = m.oid
WHERE r.rolname IN ('flight_manager', 'flight_analyst');
--13
GRANT SELECT ON flights, airline, booking
TO flight_analyst;
SELECT *
FROM information_schema.table_privileges
WHERE grantee = 'flight_analyst';
--14
GRANT SELECT ON ALL TABLES IN SCHEMA public
TO flight_manager;
GRANT INSERT, UPDATE ON booking
TO flight_manager;
GRANT USAGE, SELECT ON ALL SEQUENCES IN SCHEMA public
TO flight_manager;
Data Output Messages Notifications
GRANT
Query returned successfully in 160 msec.
Query returned successfully in 160 msec.
Total rows:
Query complete 00:00:00.160
CRLF
Ln 242, Col 1

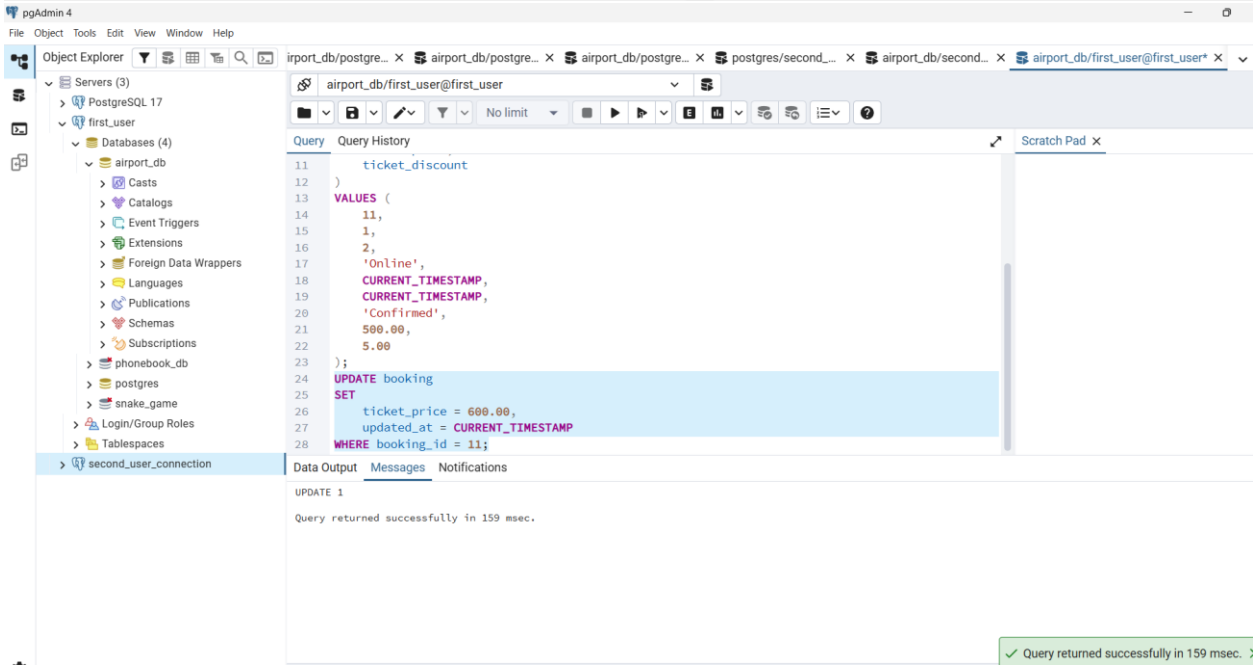

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
--13
GRANT SELECT ON flights, airline, booking
TO flight_analyst;
SELECT *
FROM information_schema.table_privileges
WHERE grantee = 'flight_analyst';
--14
GRANT SELECT ON ALL TABLES IN SCHEMA public
TO flight_manager;
GRANT INSERT, UPDATE ON booking
TO flight_manager;
GRANT USAGE, SELECT ON ALL SEQUENCES IN SCHEMA public
TO flight_manager;
SELECT *
FROM information_schema.table_privileges
WHERE grantee = 'flight_manager';
Data Output Messages Notifications
Showing rows: 1 to 19
Page No: 1 of 1
grantor grantee table_catalog table_schema table_name privilege_type is_grantable with_hierarchy
1 postgres flight_manag... airport_db public flights SELECT NO YES
2 postgres flight_manag... airport_db public airline SELECT NO YES
3 postgres flight_manag... airport_db public baggage SELECT NO YES
4 postgres flight_manag... airport_db public airport SELECT NO YES
5 postgres flight_manag... airport_db public passengers SELECT NO YES
6 postgres flight_manag... airport_db public passenger_3nf SELECT NO YES
Total rows: 19
Query complete 00:00:00.245
CRLF
Ln 248, Col 1

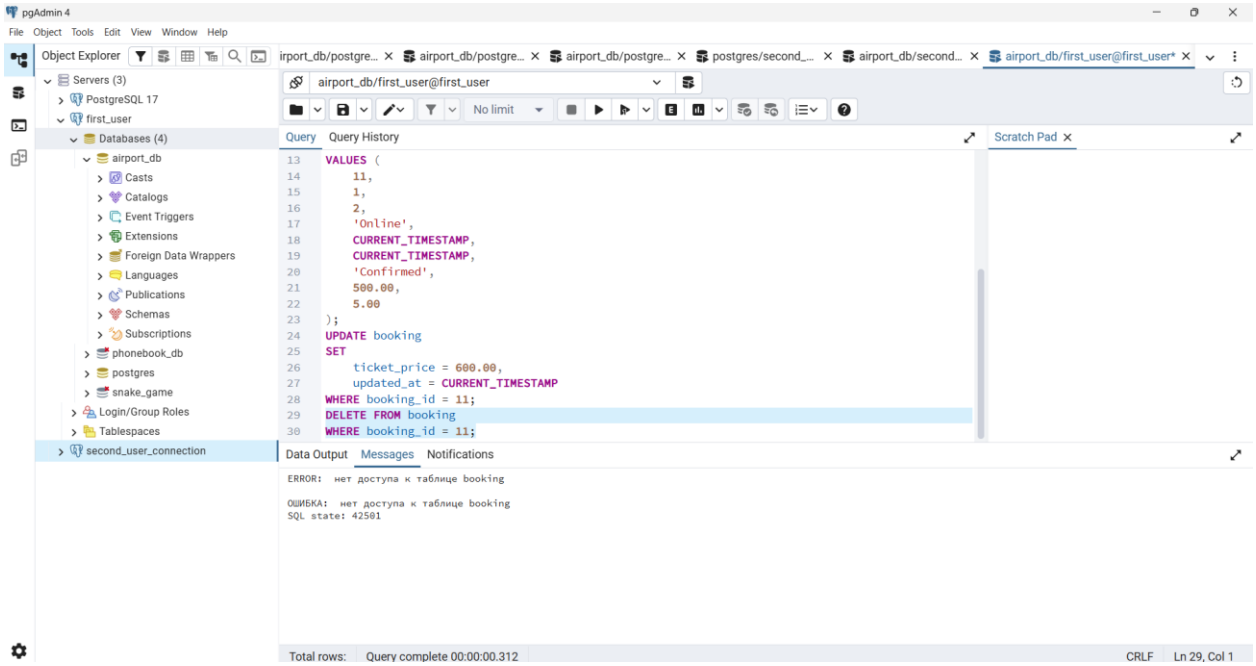
15


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
Servers (2)
PostgreSQL 17
second_user_connection
Databases (4)
airport_db
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas
Subscriptions
phonebook_db
postgres
snake_game
Login/Group Roles
Tablespaces
airport_db/second_user@second_user_connection
No limit
Query
Query History
Scratch Pad
--15
SELECT * FROM flights;
DELETE FROM booking
WHERE booking_id = 1;
Data Output
Messages
Notifications
ERROR: нет доступа к таблице booking
ОШИБКА: нет доступа к таблице booking
SQL state: 42501
Total rows: 5
Query complete 00:00:00.256
CRLF
Ln 4, Col 22

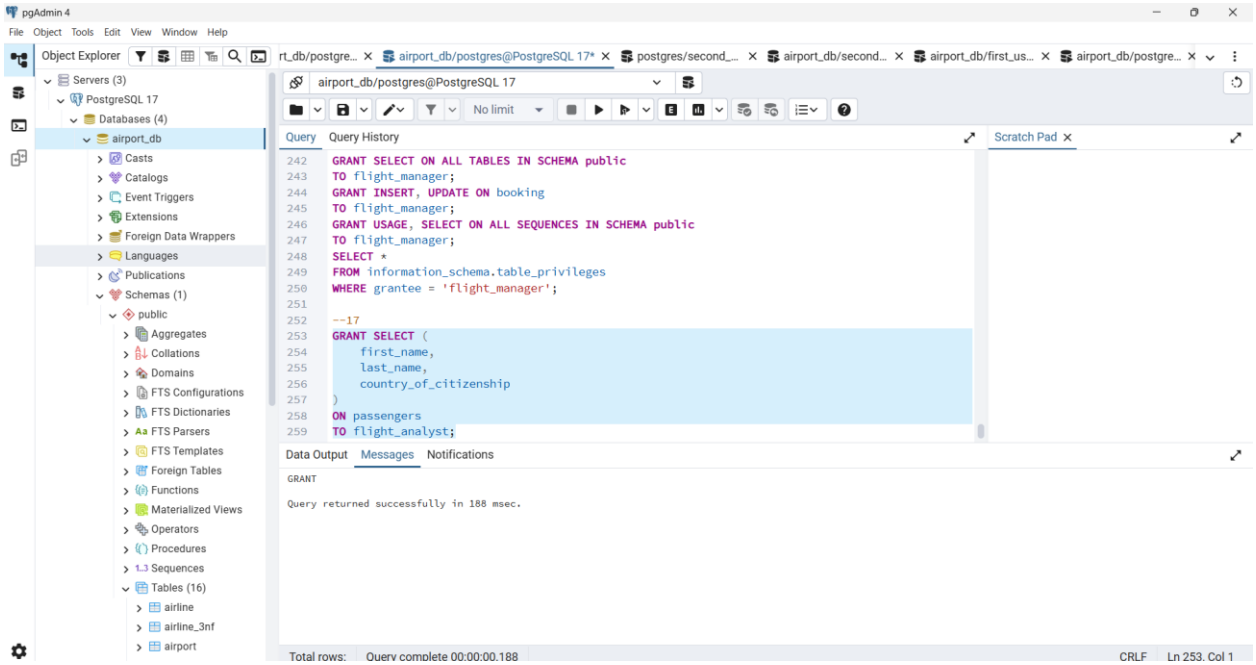
16

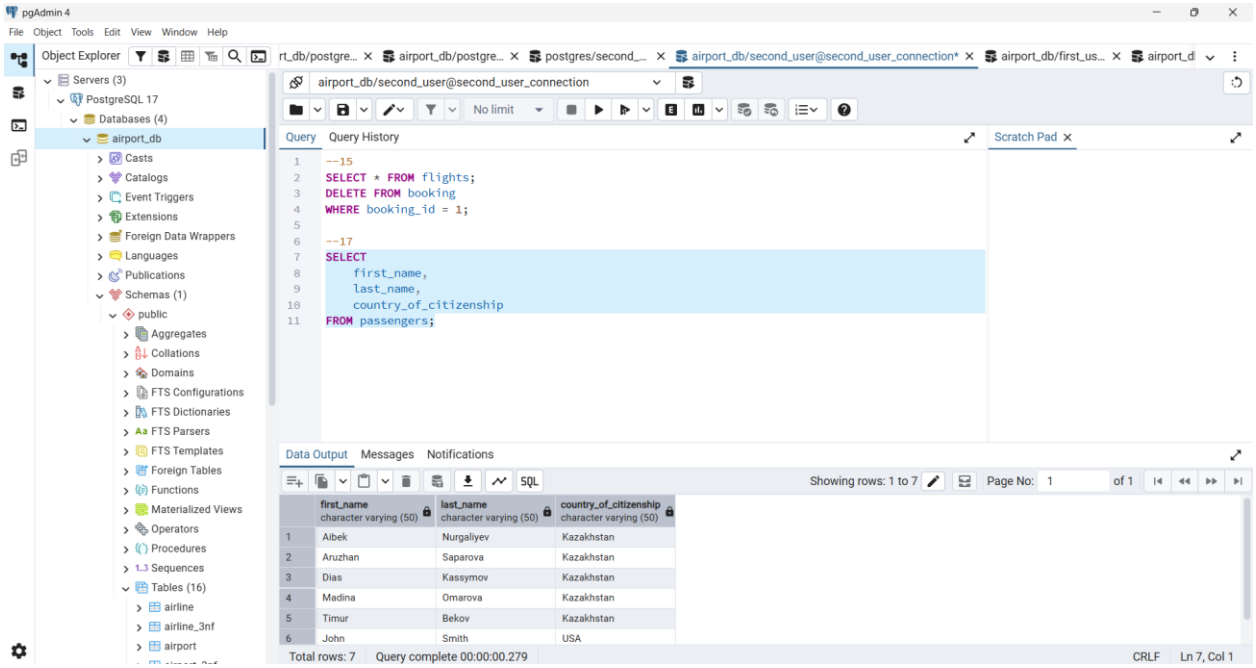

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
Servers (3)
PostgreSQL 17
first_user
Databases (4)
airport_db
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas
Subscriptions
phonebook_db
postgres
snake_game
Login/Group Roles
Tablespaces
second_user_connection
airport_db/first_user@first_user
No limit
Query
Query History
Scratch Pad
--16
INSERT INTO booking (
booking_id,
flight_id,
passenger_id,
booking_platform,
created_at,
updated_at,
status,
ticket_price,
ticket_discount
)
VALUES (
11,
1,
2,
'Online',
CURRENT_TIMESTAMP,
Data Output
Messages
Notifications
INSERT 0 1
Query returned successfully in 213 msec.
Total rows:
Query complete 00:00:00.213
CRLF
Ln 19, Col 23

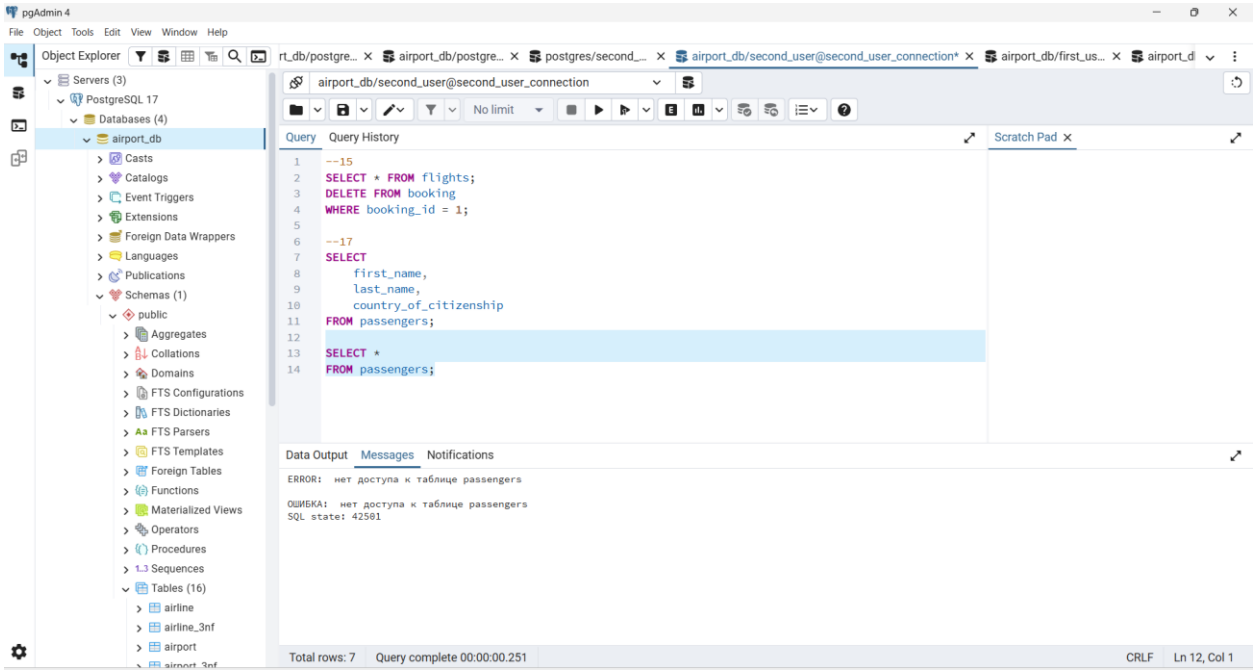

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
Servers (3)
PostgreSQL 17
first_user
Databases (4)
airport_db
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas
Subscriptions
phonebook_db
postgres
snake_game
Login/Group Roles
Tablespaces
second_user_connection
airport_db/first_user@first_user
No limit
Query
Query History
Scratch Pad
ticket_discount
)
VALUES (
11,
1,
2,
'Online',
CURRENT_TIMESTAMP,
CURRENT_TIMESTAMP,
'Confirmed',
500.00,
5.00
);
UPDATE booking
SET
ticket_price = 600.00,
updated_at = CURRENT_TIMESTAMP
WHERE booking_id = 11;
Data Output
Messages
Notifications
UPDATE 1
Query returned successfully in 159 msec.
Query returned successfully in 159 msec.


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/first_user@first_user
Query Query History
Scratch Pad
VALUES (
11,
1,
2,
'Online',
CURRENT_TIMESTAMP,
CURRENT_TIMESTAMP,
'Confirmed',
500.00,
5.00
);
UPDATE booking
SET
ticket_price = 600.00,
updated_at = CURRENT_TIMESTAMP
WHERE booking_id = 11;
DELETE FROM booking
WHERE booking_id = 11;
Data Output Messages Notifications
ERROR: нет доступа к таблице booking
ОШИБКА: нет доступа к таблице booking
SQL state: 42501
Total rows: Query complete 00:00:00.312
CRLF Ln 29, Col 1

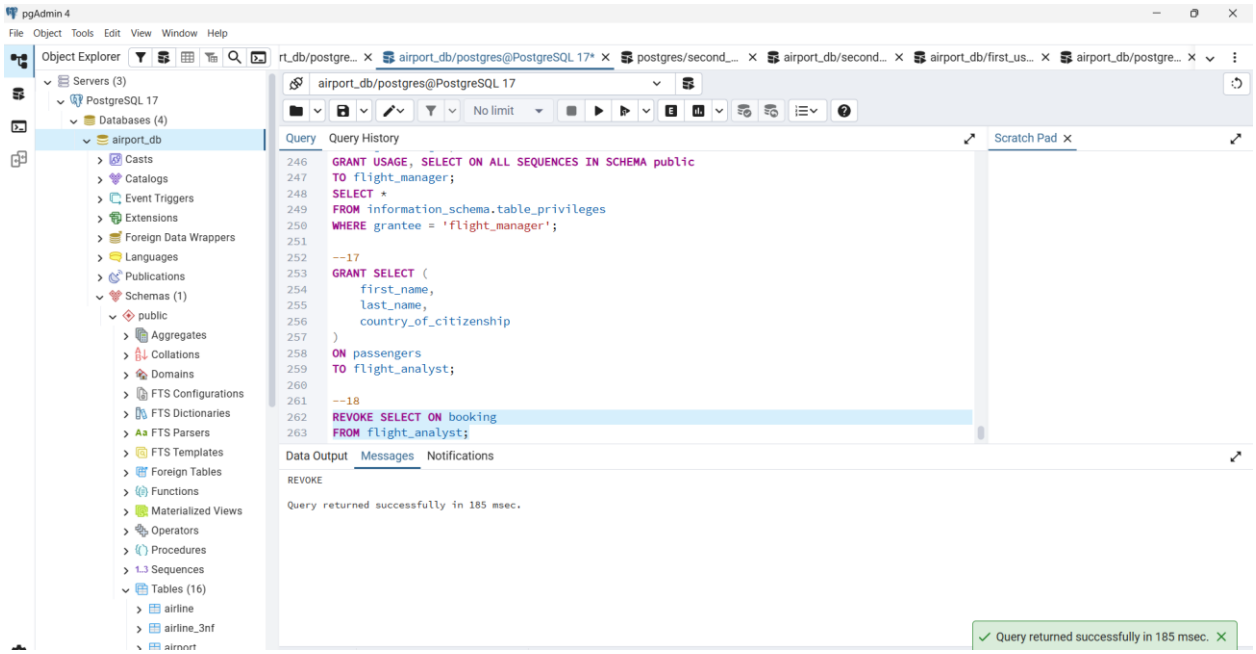
17

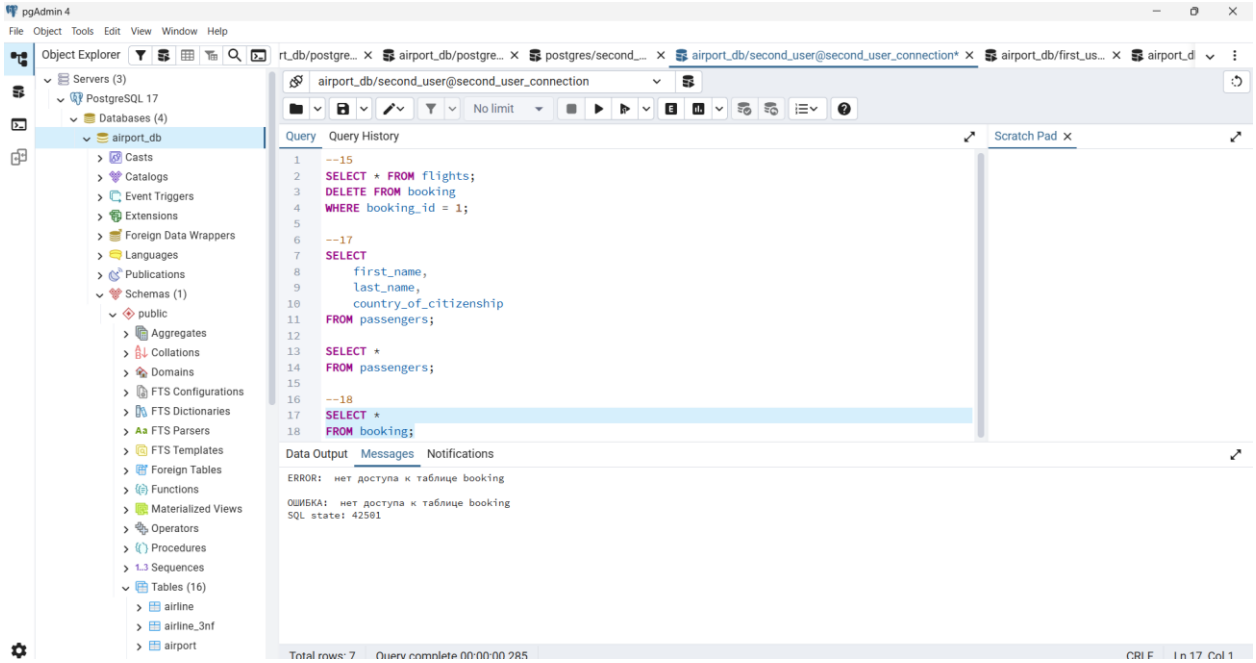

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
GRANT SELECT ON ALL TABLES IN SCHEMA public
TO flight_manager;
GRANT INSERT, UPDATE ON booking
TO flight_manager;
GRANT USAGE, SELECT ON ALL SEQUENCES IN SCHEMA public
TO flight_manager;
SELECT *
FROM information_schema.table_privileges
WHERE grantee = 'flight_manager';
--17
GRANT SELECT (
first_name,
last_name,
country_of_citizenship
)
ON passengers
TO flight_analyst;
Data Output Messages Notifications
GRANT
Query returned successfully in 188 msec.
Total rows: Query complete 00:00:00.188
CRLF Ln 253, Col 1


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/second_user@second_user_connection
No limit
Query Query History
Scratch Pad
--15
SELECT * FROM flights;
DELETE FROM booking
WHERE booking_id = 1;
--17
SELECT
first_name,
last_name,
country_of_citizenship
FROM passengers;
Data Output Messages Notifications
Showing rows: 1 to 7
Page No: 1 of 1
first_name character varying (50)
last_name character varying (50)
country_of_citizenship character varying (50)
1 Aibek Nurgaliyev Kazakhstan
2 Aruzhan Saparova Kazakhstan
3 Dias Kassymov Kazakhstan
4 Madina Omarova Kazakhstan
5 Timur Bekov Kazakhstan
6 John Smith USA
Total rows: 7 Query complete 00:00:00.279
CRLF Ln 7, Col 1

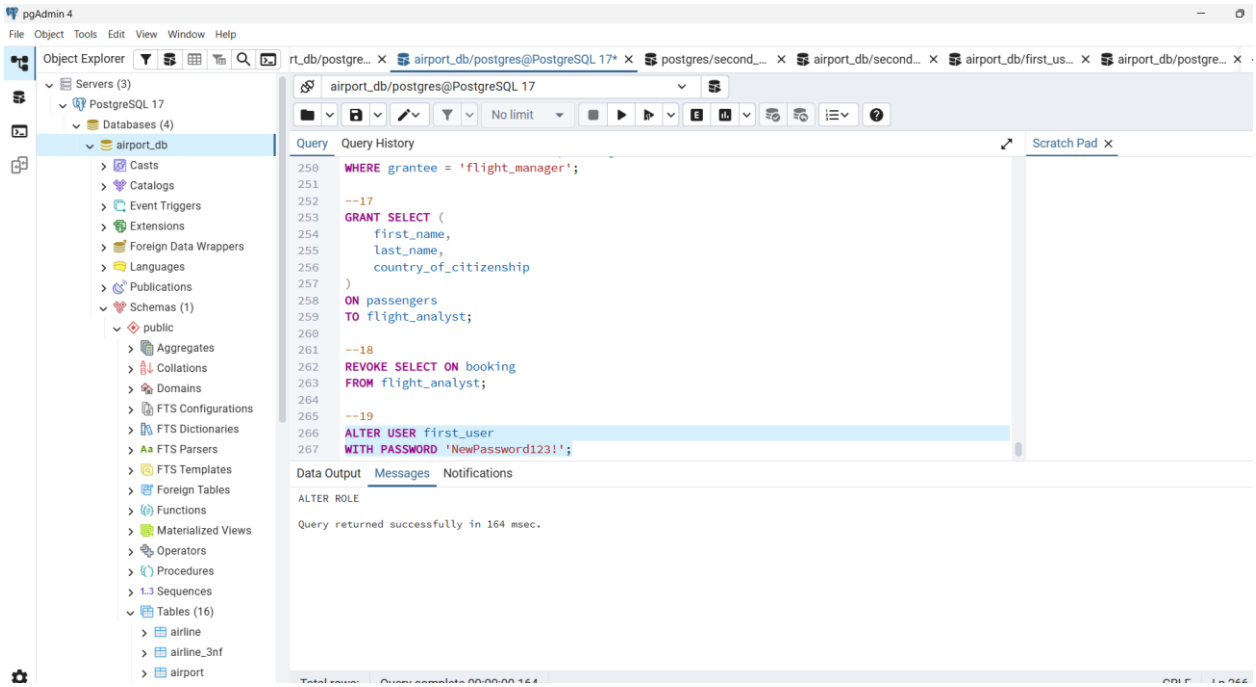

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Object Explorer
airport_db/second_user@second_user_connection
No limit
Query Query History
Scratch Pad
--15
SELECT * FROM flights;
DELETE FROM booking
WHERE booking_id = 1;
--17
SELECT
first_name,
last_name,
country_of_citizenship
FROM passengers;
SELECT *
FROM passengers;
Data Output Messages Notifications
ERROR: нет доступа к таблице passengers
ОШИБКА: нет доступа к таблице passengers
SQL state: 42501
Total rows: 7 Query complete 00:00:00.251
CRLF Ln 12, Col 1

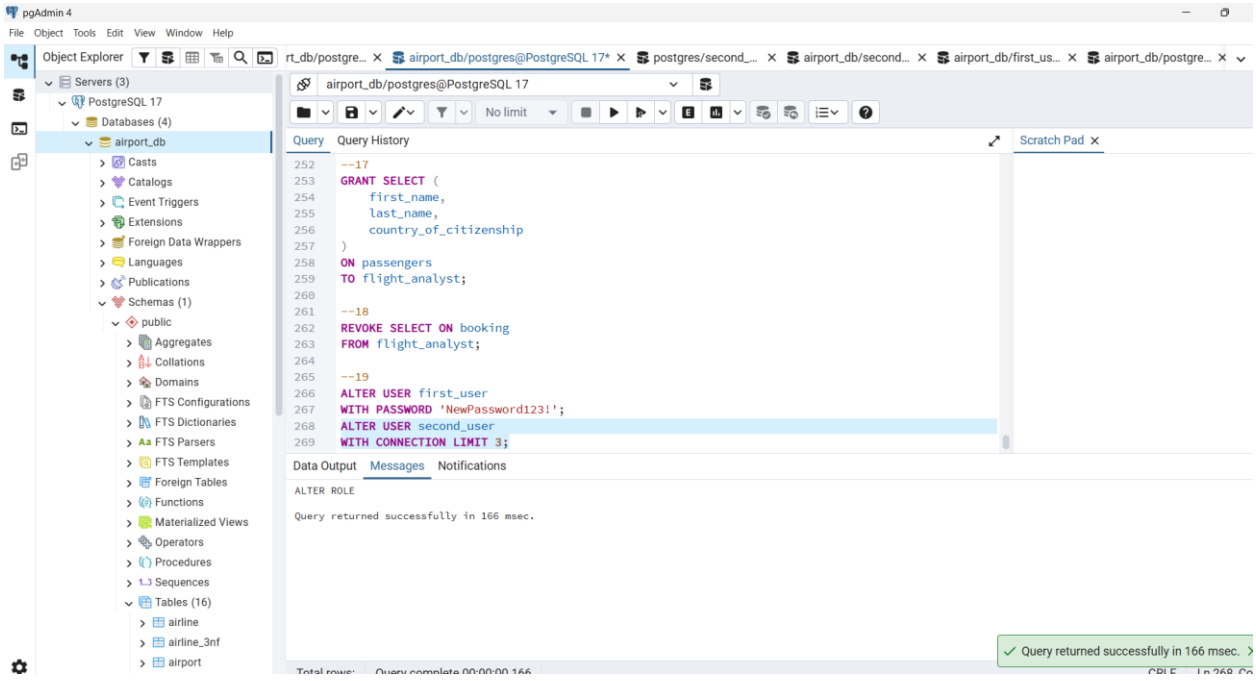
18

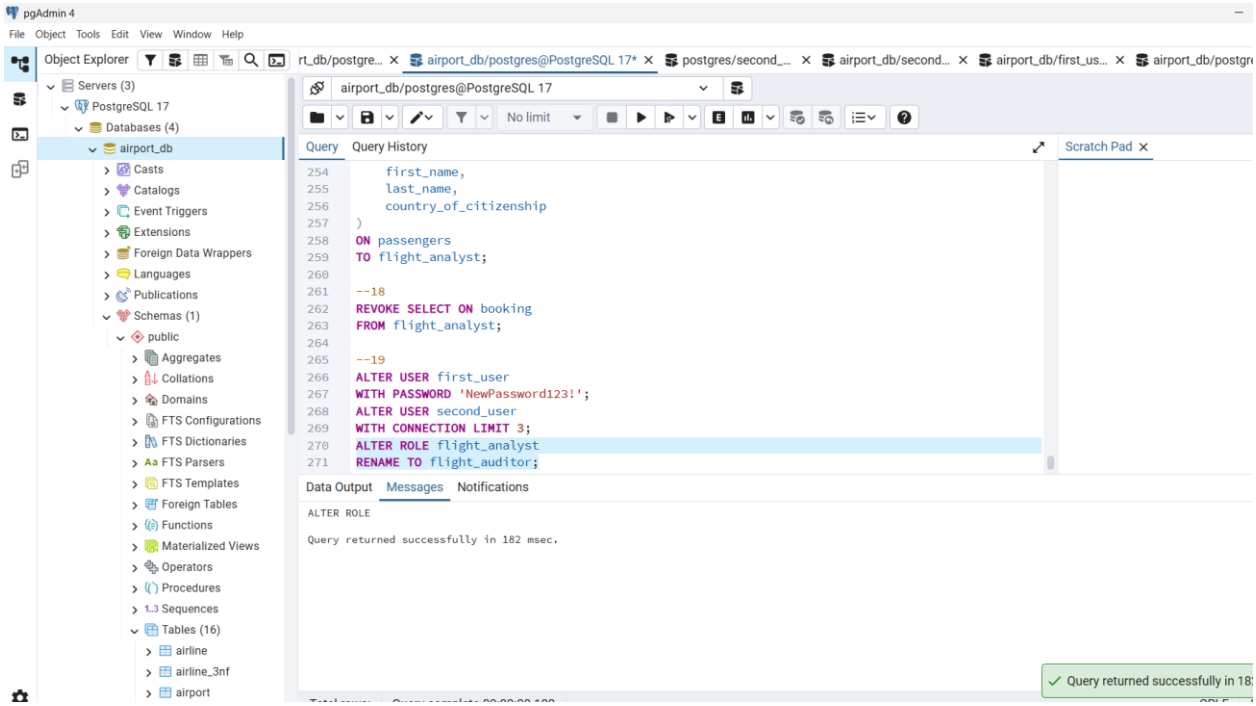

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Tables (16)
airline
airline_3nf
airport
Query
Query History
Scratch Pad
246 GRANT USAGE, SELECT ON ALL SEQUENCES IN SCHEMA public
247 TO flight_manager;
248 SELECT *
249 FROM information_schema.table_privileges
250 WHERE grantee = 'flight_manager';
251
252 --17
253 GRANT SELECT (
254 first_name,
255 last_name,
256 country_of_citizenship
257)
258 ON passengers
259 TO flight_analyst;
260
261 --18
262 REVOKE SELECT ON booking
263 FROM flight_analyst;
Data Output
Messages
Notifications
REVOKE
Query returned successfully in 185 msec.
Query returned successfully in 185 msec.

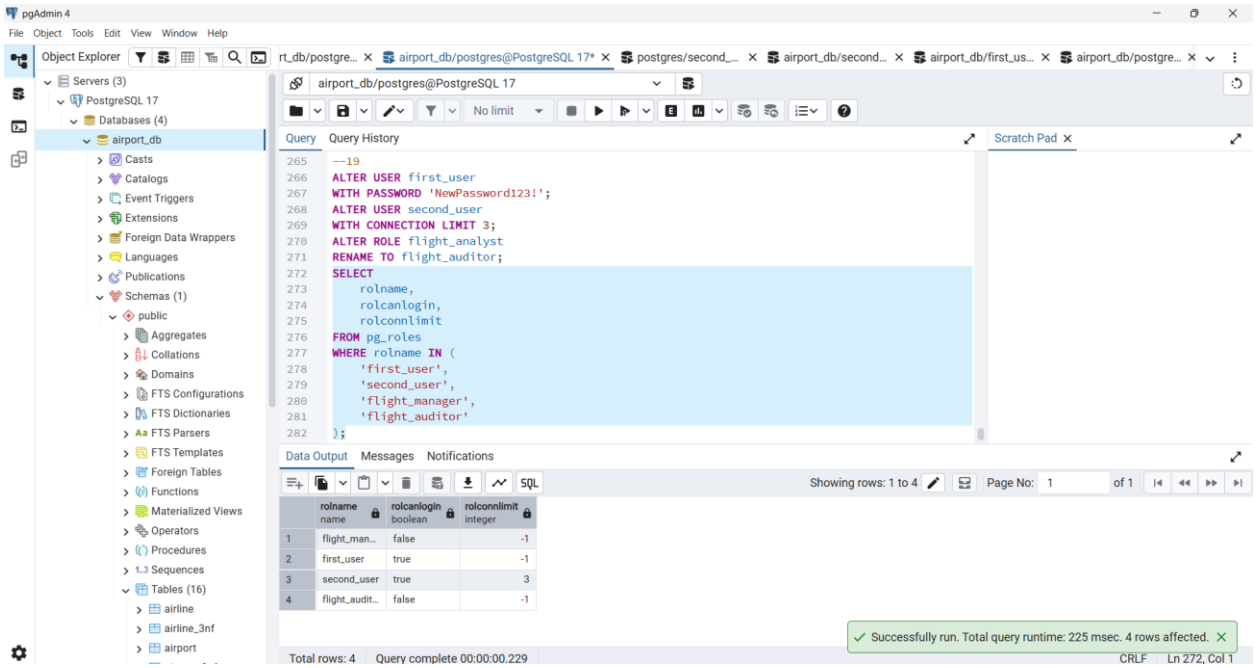

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Tables (16)
airline
airline_3nf
airport
Query
Query History
Scratch Pad
1 --15
2 SELECT * FROM flights;
3 DELETE FROM booking
4 WHERE booking_id = 1;
5
6 --17
7 SELECT
8 first_name,
9 last_name,
10 country_of_citizenship
11 FROM passengers;
12
13 SELECT *
14 FROM passengers;
15
16 --18
17 SELECT *
18 FROM booking;
Data Output
Messages
Notifications
ERROR: нет доступа к таблице booking
ОШИБКА: нет доступа к таблице booking
SQL state: 42501
Total rows: 7
Query complete 00:00:00.285
CRLF
Ln 17, Col 1

19

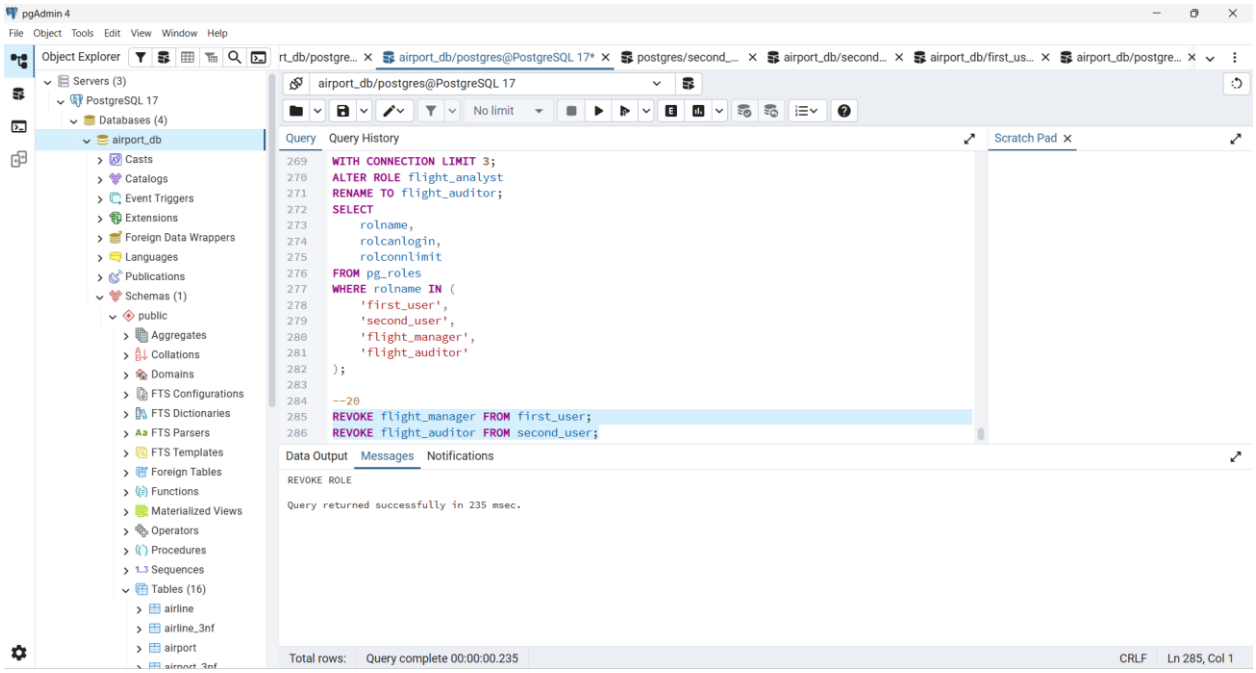

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Tables (16)
airline
airline_3nf
airport
rt_db/postgre...
airport_db/postgres@PostgreSQL 17*
postgres/second_...
airport_db/second...
airport_db/first_us...
airport_db/postgre...
airport_db/postgres@PostgreSQL 17
No limit
Query
Query History
Scratch Pad
250 WHERE grantee = 'flight_manager';
251
252 --17
253 GRANT SELECT (
254 first_name,
255 last_name,
256 country_of_citizenship
257)
258 ON passengers
259 TO flight_analyst;
260
261 --18
262 REVOKE SELECT ON booking
263 FROM flight_analyst;
264
265 --19
266 ALTER USER first_user
267 WITH PASSWORD 'NewPassword123!';
Data Output
Messages
Notifications
ALTER ROLE
Query returned successfully in 164 msec.

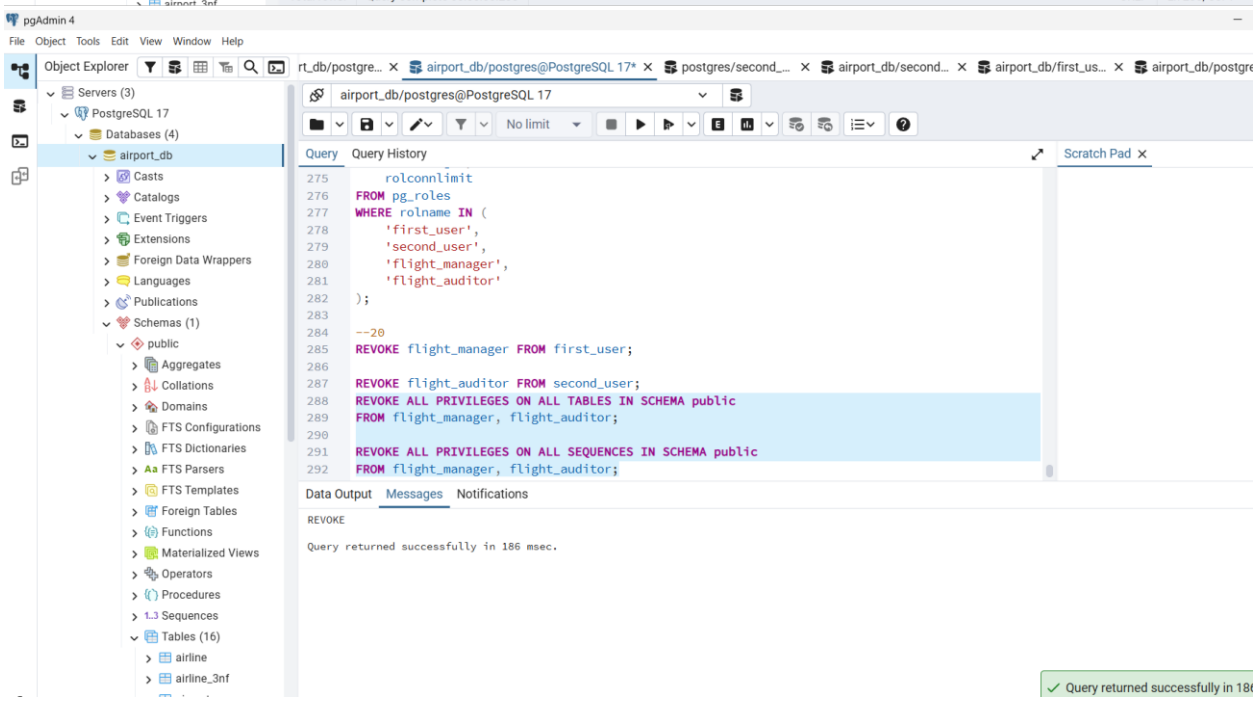

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airline_3nf
airport
rt_db/postgre...
airport_db/postgres@PostgreSQL 17*
postgres/second_...
airport_db/second...
airport_db/first_us...
airport_db/postgre...
airport_db/postgres@PostgreSQL 17
No limit
Query
Query History
Scratch Pad
252 --17
253 GRANT SELECT (
254 first_name,
255 last_name,
256 country_of_citizenship
257)
258 ON passengers
259 TO flight_analyst;
260
261 --18
262 REVOKE SELECT ON booking
263 FROM flight_analyst;
264
265 --19
266 ALTER USER first_user
267 WITH PASSWORD 'NewPassword123!';
268 ALTER USER second_user
269 WITH CONNECTION LIMIT 3;
Data Output
Messages
Notifications
ALTER ROLE
Query returned successfully in 166 msec.
Query returned successfully in 166 msec.

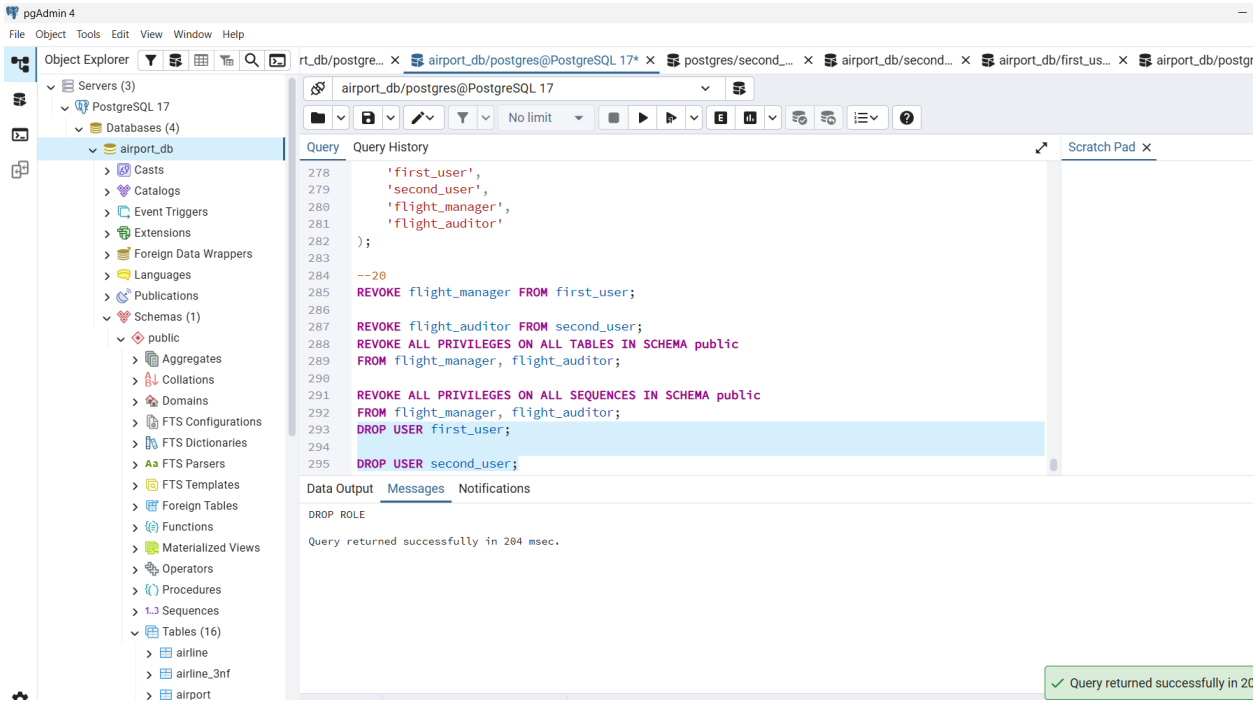

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airline_3nf
airport
No limit
Query
Query History
Scratch Pad
254 first_name,
255 last_name,
256 country_of_citizenship
257)
258 ON passengers
259 TO flight_analyst;
260
261 --18
262 REVOKE SELECT ON booking
263 FROM flight_analyst;
264
265 --19
266 ALTER USER first_user
267 WITH PASSWORD 'NewPassword123!';
268 ALTER USER second_user
269 WITH CONNECTION LIMIT 3;
270 ALTER ROLE flight_analyst
271 RENAME TO flight_auditor;
Data Output
Messages
Notifications
ALTER ROLE
Query returned successfully in 182 msec.
Query returned successfully in 18

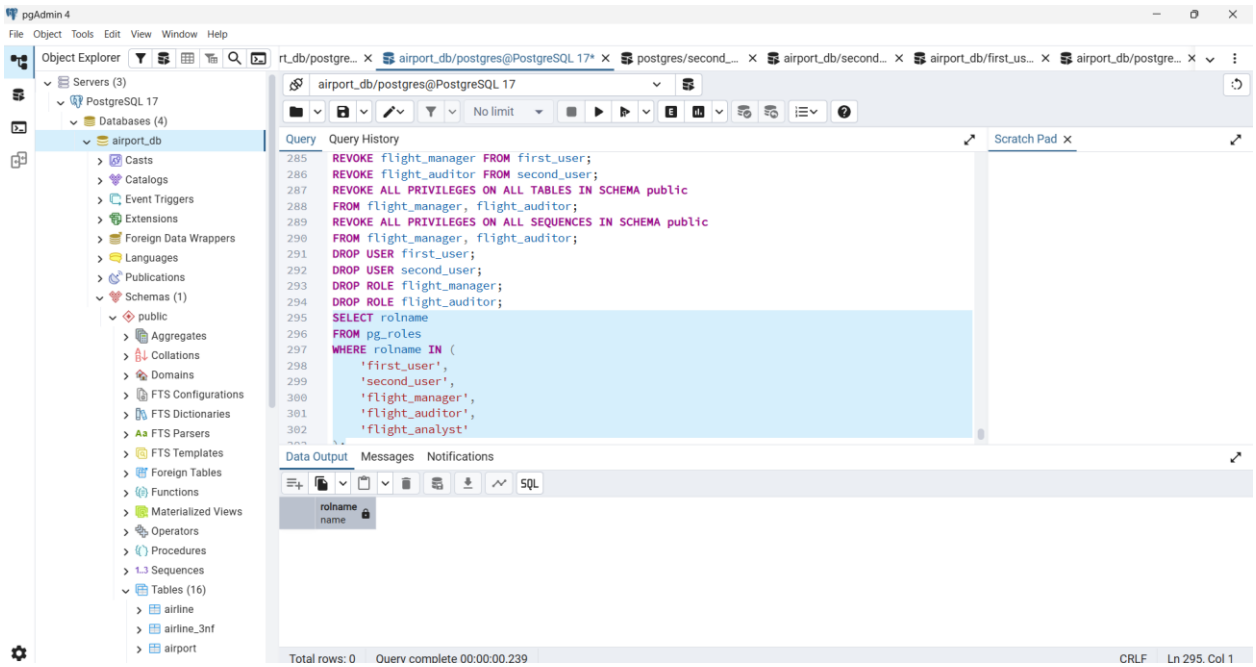

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airline_3nf
airport
No limit
Query
Query History
Scratch Pad
265 --19
266 ALTER USER first_user
267 WITH PASSWORD 'NewPassword123!';
268 ALTER USER second_user
269 WITH CONNECTION LIMIT 3;
270 ALTER ROLE flight_analyst
271 RENAME TO flight_auditor;
272 SELECT
273 rolname,
274 rolcanlogin,
275 rolconnlimit
276 FROM pg_roles
277 WHERE rolname IN (
278 'first_user',
279 'second_user',
280 'flight_manager',
281 'flight_auditor'
282);
Data Output
Messages
Notifications
Showing rows: 1 to 4
Page No: 1
of 1
rolname name
rolcanlogin boolean
rolconnlimit integer
1 flight_man... false -1
2 first_user true -1
3 second_user true 3
4 flight_audit... false -1
Successfully run. Total query runtime: 225 msec. 4 rows affected.
Total rows: 4
Query complete 00:00:00.229
CRLF
Ln 272, Col 1

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airport_db/postgres@PostgreSQL 17*
Servers (3)
PostgreSQL 17
Databases (4)
airport_db
Query Query History
Scratch Pad
WITH CONNECTION LIMIT 3;
ALTER ROLE flight_analyst
RENAME TO flight_auditor;
SELECT
rolname,
rolcanlogin,
rolconnlimit
FROM pg_roles
WHERE rolname IN (
'first_user',
'second_user',
'flight_manager',
'flight_auditor'
);
--20
REVOKE flight_manager FROM first_user;
REVOKE flight_auditor FROM second_user;
Data Output Messages Notifications
REVOKE ROLE
Query returned successfully in 235 msec.
Total rows: Query complete 00:00:00.235
CRLF Ln 285, Col 1


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Databases (4)
airport_db
Query Query History
Scratch Pad
rolconnlimit
FROM pg_roles
WHERE rolname IN (
'first_user',
'second_user',
'flight_manager',
'flight_auditor'
);
--20
REVOKE flight_manager FROM first_user;
REVOKE flight_auditor FROM second_user;
REVOKE ALL PRIVILEGES ON ALL TABLES IN SCHEMA public
FROM flight_manager, flight_auditor;
REVOKE ALL PRIVILEGES ON ALL SEQUENCES IN SCHEMA public
FROM flight_manager, flight_auditor;
Data Output Messages Notifications
REVOKE
Query returned successfully in 186 msec.
Query returned successfully in 18


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airline_3nf
airport
No limit
Query
Query History
Scratch Pad
'first_user',
'second_user',
'flight_manager',
'flight_auditor'
);
--20
REVOKE flight_manager FROM first_user;
REVOKE flight_auditor FROM second_user;
REVOKE ALL PRIVILEGES ON ALL TABLES IN SCHEMA public
FROM flight_manager, flight_auditor;
REVOKE ALL PRIVILEGES ON ALL SEQUENCES IN SCHEMA public
FROM flight_manager, flight_auditor;
DROP USER first_user;
DROP USER second_user;
Data Output
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Notifications
DROP ROLE
Query returned successfully in 204 msec.
Query returned successfully in 20


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airport
No limit
Query
Query History
Scratch Pad
REVOKE flight_manager FROM first_user;
REVOKE flight_auditor FROM second_user;
REVOKE ALL PRIVILEGES ON ALL TABLES IN SCHEMA public
FROM flight_manager, flight_auditor;
REVOKE ALL PRIVILEGES ON ALL SEQUENCES IN SCHEMA public
FROM flight_manager, flight_auditor;
DROP USER first_user;
DROP USER second_user;
DROP ROLE flight_manager;
DROP ROLE flight_auditor;
SELECT rolname
FROM pg_roles
WHERE rolname IN (
'first_user',
'second_user',
'flight_manager',
'flight_auditor',
'flight_analyst'
Data Output
Messages
Notifications
SQL
rolname
name
Total rows: 0
Query complete 00:00:00.239
CRLF
Ln 295, Col 1